

Benchmarking post-processing methods in local differential privacy for utility and adversarial robustness[☆]

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ABSTRACT

Local differential privacy (LDP) has recently gained prominence as a powerful paradigm for collecting and analyzing sensitive data from users' devices. However, the inherent perturbation added by LDP protocols reduces the utility of the collected data. To mitigate this issue, several post-processing (PP) methods have been developed. Yet, the comparative performance of PP methods under diverse utility and adversarial settings remains underexplored. In this paper, we present LDP³⁺, an open-source and extensible benchmark platform for analyzing LDP protocols and PP methods under both utility and adversarial perspectives. The platform integrates six widely used LDP protocols, seven PP methods, three adversarial attacks, multiple utility and adversarial evaluation metrics, and a multi-threaded execution framework for scalable experimentation. Using LDP³⁺, we perform extensive experiments on several real-world and synthetic datasets. Our results reveal that the effectiveness of PP methods depends on factors such as the privacy budget, protocol choice, dataset characteristics, and evaluation metric. While Norm-Sub and Norm-Mul generally provide strong utility improvements for frequency estimation, Norm-Cut performs best for rank preservation tasks. Furthermore, we show that PP methods can indirectly improve robustness against adversarial attacks such as poisoning and budget inference attacks. Overall, our work offers new insights into the role of PP methods towards improving utility and adversarial robustness, and provides an open-source benchmarking platform to support future research and deployment.

1. Introduction

Over the last few years, local differential privacy (LDP) has become a popular notion for privacy-preserving data collection and analysis. In LDP, users perturb their sensitive data locally and then send it to a data collector. The added perturbation disables the data collector from making inferences regarding any particular user, but the data collector can still estimate population-level statistics from aggregates of perturbed data. This feature of LDP makes it well-suited for data collection in decentralized settings. Consequently, LDP has seen widespread adoption in various research fields (Cormode et al., 2018; Du et al., 2023; Gursoy et al., 2019; Kim et al., 2019; Yang et al., 2023) as well as industry products, e.g., Google Chrome, Apple iOS, and Microsoft Windows (Ding et al., 2017; Erlingsson et al., 2014; Thakurta et al., 2017).

With the emerging popularity of LDP, numerous LDP protocols have been developed in the literature (Cormode et al. (2021), Gursoy et al. (2022) and Wang et al. (2017)). However, since these protocols

inherently introduce perturbation to ensure privacy, the population-level statistics estimated by the data collector become noisy, leading to utility loss. To reduce utility loss, several post-processing (PP) methods were proposed (Jia and Gong, 2019; Wang et al., 2020), e.g., Base-Pos, Norm, Norm-Cut, Norm-Sub, Norm-Mul, Power, and Power-NS. In essence, PP methods take as input the estimation results and process them in different ways to improve utility. Although PP methods have been utilized in specific contexts or tested with specific protocols, to the best of our knowledge, there is no benchmark or detailed analysis of how different PP methods behave comparatively under diverse settings, e.g., under different combinations of LDP protocols, data characteristics, utility metrics, and adversarial attacks.

To address this need, we design and develop LDP³⁺ (pronounced LDP-Cube-Plus), an open-source and modular benchmark platform consisting of six modules. Its LDP Protocol Module contains six state-of-the-art LDP protocols (GRR, RAPPOR, OUE, BLH, OLH, SS), its Post-Processing Module contains seven PP methods (Base-Pos, Norm, Norm-Cut, Norm-Sub, Norm-Mul, Power, Power-NS), its Adversarial

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Module contains three attacks (MGA, BIA, and Longitudinal attacks), and its Metrics Module contains three adversarial metrics (GR, BIE, LASR) and five utility metrics (ℓ_1 , ℓ_2 , EMD, KL-divergence, Kendall τ). The design of LDP³⁺ is modular and extensible so that new protocols, PP methods, metrics, and attacks can be added in the future. Furthermore, LDP³⁺ supports multi-threaded execution (through its Multi-Threading Module) to reduce execution times, enabling researchers and practitioners to perform fast benchmarking of protocols and PP methods. This way, we intend LDP³⁺ to serve as a practical system that helps a practitioner choose the best LDP protocol and PP method for their setting, e.g., for their desired dataset, privacy budget, and utility metric.

We perform experimental analyses using LDP³⁺ and report (a subset) of our results in the paper. Our results and analyses yield several interesting findings related to utility. First, we find that PP methods indeed yield substantial utility benefits when the privacy budget ϵ is small (strict privacy), but their utility results converge as ϵ becomes larger, and their benefits decrease. Second, we find that the best-performing PP method (in terms of utility improvement) is not static, but rather, it depends on a variety of factors including the LDP protocol, ϵ budget, data characteristics (e.g., statistical distribution, domain size), and utility metric. Third, we find that the best-performing PP method is more consistent when ϵ is small, but it becomes more inconsistent as ϵ increases. Finally, we find that Norm-Sub and Norm-Mul emerge as strong choices as the PP method to use in frequency estimation tasks, whereas Norm-Cut emerges as the best choice of PP method in rank estimation tasks.

Beyond utility, we also analyze the behavior of PP methods under adversarial attacks, which is a perspective that has not been explored in prior works. We study three attack types: (i) Maximal Gain Attack (MGA), a poisoning attack in which malicious users inject crafted reports to boost the frequencies of targeted items; (ii) Budget Inference Attack (BIA), an inference attack in which the attacker attempts to infer the true privacy budget ϵ from the collected perturbed reports; and (iii) longitudinal attacks, in which an attacker exploits repeated observations of the same user across multiple collection rounds to infer their true value. Our analyses reveal that PP methods can indirectly act as “defenses” against MGA and BIA. Norm-Mul, Norm-Sub, and Power-NS achieve greatest reductions in attackers’ gains in MGA, and while a single PP method does not dominate others in BIA, Norm-Mul and Norm-Cut more reliably reduce BIA effectiveness. In contrast, for longitudinal attacks, we find that PP methods offer little protection, as these attacks rely on repeated per-user observations that post-processing does not modify. These findings reveal interesting side benefits of PP methods towards increasing robustness against attacks.

Contributions. In short, in this paper we make the following main contributions:

- We design and develop LDP³⁺, containing the implementations of six LDP protocols, seven PP methods, three attacks, three adversarial metrics, and five utility metrics. LDP³⁺ is a modular, extensible, and multi-threaded platform intended to analyze the behaviors and optimality of PP methods under diverse settings.
- We perform experimental utility analyses of PP methods using LDP³⁺, and show that the best-performing PP method depends on a variety of factors, including the LDP protocol, ϵ budget, data characteristics, and utility metric. Nevertheless, our results indicate that Norm-Sub and Norm-Mul emerge as the most desirable choices for frequency estimation tasks, whereas Norm-Cut emerges as the most desirable choice for rank estimation tasks.
- We also conduct experimental analyses to measure the impacts of PP methods on poisoning and inference attacks using LDP³⁺. We find that PP methods can indirectly act as defenses that increase robustness of protocols against MGAs and BIAs.

- To advance research in this area and to assist researchers and practitioners in selecting the best protocol and PP method for their setting, we provide an open-source implementation¹ of LDP³⁺.

Extensions to the conference version. This paper is an extension of our conference paper published in IFIP SEC 2025 (Khodaie et al., 2025). Compared to the conference version, the main extensions and enhancements are as follows. First, we added a new module to LDP³⁺: the Adversarial Module. We performed the implementations of three attacks (MGA, BIA, and longitudinal attacks) in a way that supports all LDP protocols and PP methods. Second, we significantly enhanced the Metrics Module of LDP³⁺. We added the rank preservation task as a utility task, and added the Kendall τ metric to measure its effectiveness. We also added three new adversarial metrics: GR, BIE, and LASR. Third, we enhanced the Execution Module of LDP³⁺ to support all utility and adversarial tasks. We also added support for new execution commands (for the new tasks) and YAML-based configuration files. Fourth, we performed experiments and analyses with the newly added rank preservation task. We found that PP methods, particularly Norm-Cut, usually improve rank preservation compared to raw LDP estimates, especially in low to mid-range ϵ values. Fifth, we performed experiments and analyses with the newly added MGAs, BIAs, and longitudinal attacks. We found that PP methods are also beneficial to protect against BIAs and MGAs. We found that Norm-Mul is particularly effective against MGA, and while a single PP method cannot be singled out as optimal for BIAs, Norm-Mul and Norm-Cut are the most consistent in improving robustness against BIAs. Finally, we added more complete descriptions of many sections and LDP³⁺ modules that we could not fully include in our previous conference paper due to the page limit, e.g.: Related Work (Section 2), and the Protocol Module, Multi-Threading Module, and Execution Module of LDP³⁺.

2. Related work

2.1. LDP protocols and post-processing methods

LDP has recently been applied in a variety of contexts and data analysis tasks, such as multi-dimensional data analysis (da Costa Filho and Machado, 2023; Arcolezi et al., 2021; Zhang et al., 2018), set-valued data analysis (S. Wang et al., 2023; Y. Huang et al., 2024), location data analysis (Hong et al., 2022; Du et al., 2023; Alptekin et al., 2025), graph data analysis (Imola et al., 2021; Lin et al., 2022), and deep learning (B. Wang et al., 2023; Truex et al., 2020). Many of these applications treat well-known LDP protocols (such as GRR, OUE, and OLH) as building blocks and propose extensions and adaptations tailored to the data type and data analysis task at hand. RAPPOR and GRR are among the first LDP protocols – RAPPOR was developed by Google (Erlingsson et al., 2014) and GRR is the generalized form of the randomized response primitive used in survey data collection (Warner, 1965; Duchi et al., 2013; Kairouz et al., 2016). SS was proposed and developed in Wang et al. (2016) and Ye and Barg (2018). OLH and OUE were proposed in Wang et al. (2017), and they were shown to reduce estimation variance compared to prior protocols.

It was also shown that estimation utility in LDP can be improved using post-processing (PP) methods. Wang et al. (2020) introduced consistency-enforcing PP methods, e.g., correcting negative frequency estimates to zero or normalizing estimates so that they sum to 1. To address estimation errors in small-sample or high-privacy regimes, Murakami et al. (2018) proposed an Expectation-Maximization (EM)-based reconstruction method. Ye et al. (2025) recently revisited EM-based estimation for LDP protocols. We note that these EM-based methods fall outside the scope of LDP³⁺, as we define and use PP methods as lightweight operators that are directly applied on the

¹ <https://github.com/alrzhakh/LDPCubePlus>

frequency estimate produced by a protocol's standard estimation procedure. In contrast, EM-based methods constitute alternative, iterative estimation/reconstruction procedures that replace the standard protocol estimation. Benchmarking this broader class of estimators is a valuable direction, which we leave for future work. ElSalamouny and Palamidessi [ElSalamouny and Palamidessi \(2020\)](#) introduced the Generalized Iterative Bayesian Update (GIBU), which aggregates data from users with diverse privacy mechanisms into a single maximum likelihood estimate. [Arcolezi et al. \(2023a\)](#) studied the utility gain of iterative Bayesian updates in LDP. [Li et al. \(2025\)](#) conducted an empirical analysis of PP methods for recovering correct estimates from poisoned data. Our past work (the conference version of this paper) analyzed the comparative performance of various PP methods under different settings such as different LDP protocols, utility metrics, and datasets ([Khodaie et al., 2025](#)).

2.2. Adversarial attacks on LDP protocols

The works cited above have primarily focused on improving utility, such as reducing estimation error or ensuring statistical consistency. In contrast, one of the key contributions of LDP³⁺ (also one of its distinguishing aspects beyond LDP³) is that it incorporates adversarial aspects. The intersection between PP methods and adversarial attacks remains underexplored in the literature, and LDP³⁺ addresses this gap by providing a platform that enables systematic experimental analysis of how different PP methods interact with different adversarial attacks.

The first class of attacks considered in LDP³⁺ is poisoning attacks. In poisoning attacks, malicious users poison their LDP protocol inputs or outputs to manipulate the statistics recovered on the data collector's side. The vulnerability of LDP protocols to poisoning attacks was first demonstrated by [Cheu et al. \(2021\)](#) and [Cao et al. \(2021\)](#), who showed that estimation results can be significantly distorted by attackers. Subsequent work extended poisoning attacks to other settings and tailored attacks to specific data types and applications. For example, [Li et al. \(2023\)](#) proposed poisoning strategies targeting mean and variance estimation, while [Li et al. \(2024\)](#) studied the robustness of LDP protocols for numerical data against poisoning. [Wu et al. \(2022\)](#) and [Yaguchi and Kikuchi \(2025\)](#) investigated poisoning attacks against LDP protocols designed for key-value data collection. [Tong et al. \(2024\)](#) studied poisoning attacks on frequent itemset mining protocols. [Imola et al. \(2022\)](#) studied the robustness of LDP protocols for graph analysis. [Wang et al. \(2022\)](#) proposed methods for analyzing preference data under LDP with enhanced utility and robustness, and [Zheng et al. \(2024\)](#) studied poisoning attacks against LDP-based crowdsensing.

The second class of attacks considered in LDP³⁺ is privacy-related attacks, i.e., inference attacks. These attacks aim at inferring sensitive information from LDP protocol outputs or quantifying the behaviors against specific adversary types. [Murakami and Takahashi \(2021\)](#) studied the re-identification risks in LDP protocols. [Gadotti et al. \(2022\)](#) proposed pool inference attacks in which an adversary tries to infer whether a user's true value falls within a pool of interest. Bayesian inference attacks were first proposed in [Gursoy et al. \(2022\)](#), and later extended and used in [Gursoy \(2024\)](#), [Arcolezi and Gambis \(2025\)](#) and [Balioglu et al. \(2025\)](#) for longitudinal inference and measuring protocol fairness. [Arcolezi et al. \(2023b\)](#) studied the risks stemming from multi-dimensional data collection with LDP. Budget inference attacks were proposed in [Balioglu and Gursoy \(2026\)](#), where the adversary's goal is to infer the privacy budget used in data collection.

2.3. LDP packages and toolkits

While there exist a few past works in benchmarking LDP protocols or developing open-source LDP platforms, our LDP³⁺ has several key differences and advantages. Earlier works such as [Wang et al. \(2017\)](#) and [Wang et al. \(2020\)](#) focus solely on protocols without post-processing, or they evaluate post-processing only on a single protocol.

Table 1

Summary of notations used throughout the paper.

Symbol	Description
$\mathcal{P}$	Set of users (population)
u	A user in the population, $u \in \mathcal{P}$
$\mathcal{D}$	Domain of possible values (items)
v_u	True value of user u , where $v_u \in \mathcal{D}$
$f(v)$	True frequency of v
$\hat{f}(v)$	Estimated frequency obtained from an LDP protocol (before PP)
$\tilde{f}(v)$	Post-processed frequency after applying a PP method
$\mathbf{f}$	Collection of true frequencies $\{f(v)\}_{v \in \mathcal{D}}$
$\hat{\mathbf{f}}$	Collection of estimated frequencies $\{\hat{f}(v)\}_{v \in \mathcal{D}}$
$\tilde{\mathbf{f}}$	Collection of post-processed frequencies $\{\tilde{f}(v)\}_{v \in \mathcal{D}}$
ϵ	Privacy budget in LDP
T	Target set of attacker in MGA (poisoning attack)
$g^p(v)$	Gain of a poisoning attack without post-processing
$g^{pp}(v)$	Gain of a poisoning attack after post-processing
$\mathcal{P}_{adv}$	Set of malicious users injected by the attacker in MGA
$\mathcal{O}_u$	Observed perturbed reports from user u (longitudinal attack)
v_u^{pred}	Value predicted by the attacker for user u in a longitudinal attack

PURE-LDP proposed in [Cormode et al. \(2021\)](#) contains a subset of the LDP protocols and PP methods that exist in LDP³⁺; in addition, it does not contain adversarial aspects or multi-threading. The MULTI-FREQ-LDP package was proposed in [Arcolezi et al. \(2022b\)](#); however, its focus is on multi-dimensional and longitudinal frequency estimation ([Arcolezi et al., 2021, 2022a](#)) rather than post-processing and adversarial aspects. LDPLENS was proposed in [Gursoy et al. \(2022\)](#); however, it is focused on protocols' adversarial analysis and does not contain post-processing. LDP-AUDITOR was proposed in [Arcolezi and Gambis \(2024\)](#) for auditing the correctness of protocol implementations and empirically estimating their privacy loss; however, it is not related to post-processing. LDP-TOOLBOX was proposed in [Zhang et al. \(2025\)](#) for exploring the utility and attackability trade-offs in LDP; however, it is also not related to post-processing. Overall, our work differs from prior works primarily due to its emphasis on post-processing, as well as its novel experimental analysis and take-away messages regarding post-processing methods.

3. Background and notation

We start by establishing the fundamental notations and definitions related to local differential privacy (LDP) and our context. A summary of the notations used throughout the paper is provided in [Table 1](#).

LDP is a popular notion for collecting data while protecting individuals' privacy. In a typical LDP setup, there are multiple users (clients) and a server (data collector). We denote the user population by $\mathcal{P}$, a user by $u \in \mathcal{P}$, and the user's true value by $v_u \in \mathcal{D}$, where $\mathcal{D}$ is the domain of allowed values. To satisfy LDP, each user encodes and perturbs their v_u locally on their device using a randomized algorithm ψ . The user then sends the perturbed value to the server. After gathering data from all users, the server applies estimation methods to compute statistics about the overall population $\mathcal{P}$. However, since each user's data is perturbed using ψ , the server cannot infer exact information about any particular user's v_u .

Definition 1 (ϵ -LDP). A randomized mechanism ψ satisfies ϵ -LDP if and only if, for any two values $v_1, v_2 \in \mathcal{D}$:

$$\forall y \in \text{Range}(\psi) : \frac{\Pr[\psi(v_1) = y]}{\Pr[\psi(v_2) = y]} \leq e^\epsilon \quad (1)$$

where $\text{Range}(\psi)$ denotes the set of all possible outputs of ψ .

ϵ -LDP guarantees that, given the perturbed value y , the server cannot distinguish whether the user's true value was v_1 or v_2 with a probability ratio greater than e^ϵ . The strength of privacy protection is determined by the parameter ϵ , often referred to as the *privacy budget*, where smaller ϵ indicates stronger privacy.

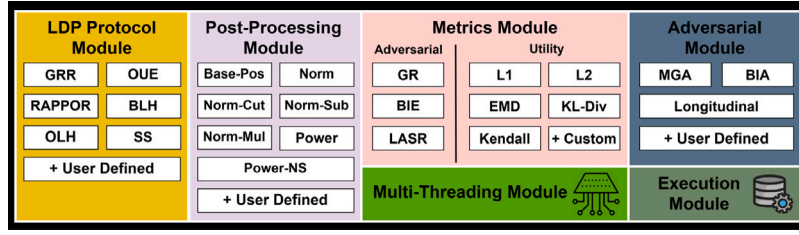


Fig. 1. Overview of the LDP³⁺ architecture consisting of six modules: the *LDP Protocol Module* (GRR, RAPPOR, OLH, OUE, BLH, SS), the *Post-Processing Module* (Base-Pos, Norm, Norm-Cut, Norm-Sub, Norm-Mul, Power, Power-NS), the *Metrics Module* (utility metrics: ℓ_1 , ℓ_2 , KL-divergence, EMD, Kendall τ ; adversarial metrics: GR, BIE, LASR), the *Adversarial Module* (MGA, BIA, Longitudinal), the *Multi-Threading Module*, and the *Execution Module*. All modules support user-defined extensions.

The popularity of LDP has led to the development of various LDP protocols (Gursoy et al., 2022; Cormode et al., 2021; Wang et al., 2017). These protocols are often used as building blocks in more complex data analysis tasks and downstream applications. For item $v \in D$, we denote the true occurrence frequency of v in the user population $\mathcal{P}$ by $f(v)$. Due to the inherent randomness and perturbation in LDP protocols, after server-side estimation, the server recovers estimated frequencies $\hat{f}(v)$. To improve utility, server-side post-processing (PP) methods take as input $\hat{f}(v)$ and produce post-processed frequencies $\tilde{f}(v)$. We describe the LDP protocols used in LDP³⁺ in Section 4.1 and the PP methods used in LDP³⁺ in Section 4.2.

4. LDP³⁺ Design and Implementation

LDP³⁺ is an open-source platform for benchmarking and analyzing different combinations of LDP protocols and post-processing methods under various data characteristics, utility metrics, and adversarial attacks. It was implemented in a modular and extensible way using the Python programming language. The design of LDP³⁺ is visualized in Fig. 1. It contains the following six modules: LDP Protocol Module, Post-Processing Module, Metrics Module, Adversarial Module, Multi-Threading Module, and Execution Module. In the rest of this section, we describe the design and current implementation of each module in more detail.

4.1. LDP protocol module

A range of LDP protocols have been proposed in the literature Cormode et al. (2021), Gursoy et al. (2022) and Wang et al. (2017), which frequently serve as fundamental building blocks within larger data analysis pipelines and downstream privacy applications. In general, an LDP protocol consists of two core procedures: (i) a user-side encoding and perturbation procedure that guarantees ϵ -LDP, and (ii) a server-side aggregation and estimation procedure that reconstructs population-level statistics from the collected noisy reports. The LDP Protocol Module of LDP³⁺ contains implementations of six popular and representative protocols: GRR, RAPPOR, OUE, BLH, OLH, and SS. The module is extensible, i.e., developers can add new protocols by implementing the protocol's user-side perturbation and server-side estimation procedures.

Generalized Randomized Response (GRR) extends the classical randomized response technique (Warner, 1965) to discrete domains of arbitrary size $|D| \geq 2$, and satisfies ϵ -LDP for any privacy parameter ϵ . For user $u \in \mathcal{P}$ holding private value $v_u \in D$, the mechanism outputs a perturbed report $y_u \in D$ according to:

$$\Pr[y_u = y] = \begin{cases} p = \frac{e^\epsilon}{e^\epsilon + |D| - 1}, & \text{if } y = v_u \\ q = \frac{1}{e^\epsilon + |D| - 1}, & \text{if } y \neq v_u \end{cases} \quad (2)$$

After collecting perturbed reports from all users in $\mathcal{P}$, to perform estimation for some value $v \in D$, the server computes: $\hat{C}(v) =$

$|\{u \in \mathcal{P} : y_u = v\}|$, which denotes the number of reports equal to v . The unbiased estimator $\hat{f}(v)$ of the true frequency $f(v)$ is then given by:

$$\hat{f}(v) = \frac{\hat{C}(v) - |P|q}{(p - q)|P|} \quad (3)$$

Binary Local Hashing (BLH) maps values into binary space (i.e., a single bit) via hashing and then perturbs the resulting bit. Let $\mathcal{H}$ denote a family of hash functions where each $H \in \mathcal{H}$ satisfies $H : D \rightarrow \{0, 1\}$. Each user $u \in \mathcal{P}$ independently samples a hash function H_u from $\mathcal{H}$ and computes: $x_u = H_u(v_u)$. The value x_u is then perturbed according to:

$$\Pr[x'_u = i] = \begin{cases} \frac{e^\epsilon}{e^\epsilon + 1}, & \text{if } i = x_u \\ \frac{1}{e^\epsilon + 1}, & \text{if } i \neq x_u \end{cases} \quad i \in \{0, 1\} \quad (4)$$

The user reports the pair $\langle H_u, x'_u \rangle$ to the server. After collecting reports from all users in $\mathcal{P}$, to perform estimation for value $v \in D$, the server computes:

$$\text{Sup}(v) = |\{u \in \mathcal{P} : x'_u = H_u(v)\}| \quad (5)$$

which corresponds to the number of reports “supporting” value v . The estimated frequency $\hat{f}(v)$ is then given by:

$$\hat{f}(v) = \frac{(e^\epsilon + 1)(2 \text{Sup}(v) - |P|)}{(e^\epsilon - 1)|P|} \quad (6)$$

Optimized Local Hashing (OLH) generalizes BLH by allowing the hash function to map domain elements into a larger output space. Instead of restricting the hash range to $\{0, 1\}$, OLH uses an integer range $[0, g - 1]$, where $g \geq 2$ is a protocol parameter. (A commonly adopted default choice is $g = e^\epsilon + 1$ (Wang et al., 2017).) This extension reduces the utility loss observed in BLH when binary hashing is suboptimal. It has been shown that OLH significantly improves estimation accuracy, particularly for moderate to large ϵ and population sizes.

Let $\mathcal{H}$ denote a family of hash functions such that each $H \in \mathcal{H}$ satisfies $H : D \rightarrow \{0, 1, \dots, g - 1\}$. Each user $u \in \mathcal{P}$ independently samples a hash function H_u from $\mathcal{H}$ and computes: $x_u = H_u(v_u)$. The value x_u is then perturbed according to:

$$\Pr[x'_u = i] = \begin{cases} \frac{e^\epsilon}{e^\epsilon + g - 1}, & \text{if } i = x_u \\ \frac{1}{e^\epsilon + g - 1}, & \text{otherwise} \end{cases} \quad (7)$$

The user reports the pair $\langle H_u, x'_u \rangle$ to the server. To perform estimation for v , the server again finds the support of v :

$$\text{Sup}(v) = |\{u \in \mathcal{P} : x'_u = H_u(v)\}| \quad (8)$$

The estimated frequency $\hat{f}(v)$ is given by:

$$\hat{f}(v) = \frac{(e^\epsilon + g - 1)(g \text{Sup}(v) - |P|)}{(e^\epsilon - 1)(g - 1)|P|} \quad (9)$$

Randomized Aggregatable Privacy-Preserving Ordinal Response (RAPPOR) achieves ϵ -LDP by first encoding the user's value into a bitvector and then applying independent randomized perturbation to

each bit. We consider the widely adopted variant of RAPPOR that utilizes unary encoding. Each user $u \in \mathcal{P}$ constructs a bitvector $B_u \in \{0, 1\}^{|D|}$ such that $B_u[v_u] = 1$ and all remaining positions of B_u are set to 0. Each bit $i \in \{1, \dots, |D|\}$ is then perturbed independently according to:

$$\Pr[B'_u[i] = 1] = \begin{cases} \frac{e^{\epsilon/2}}{e^{\epsilon/2} + 1}, & \text{if } B_u[i] = 1 \\ \frac{1}{e^{\epsilon/2} + 1}, & \text{if } B_u[i] = 0 \end{cases} \quad (10)$$

The perturbed bitvector B'_u is sent to the server. After collecting perturbed bitvectors from all users in $\mathcal{P}$, the server computes:

$$\hat{C}(v) = \sum_{u \in \mathcal{P}} B'_u[v], \quad (11)$$

which counts the total number of reported 1's at position v . The frequency estimate is then obtained as:

$$\hat{f}(v) = \frac{\hat{C}(v) + |\mathcal{P}|(\alpha - 1)}{(2\alpha - 1)|\mathcal{P}|}, \quad \text{where } \alpha = \frac{e^{\epsilon/2}}{e^{\epsilon/2} + 1} \quad (12)$$

Optimized Unary Encoding (OUE) follows the same unary encoding as RAPPOR, where each user $u \in \mathcal{P}$ constructs a bitvector $B_u \in \{0, 1\}^{|D|}$ with $B_u[v_u] = 1$ and all other positions equal to 0. The perturbation mechanism, however, adopts probabilities that are analytically derived to minimize the variance of the frequency estimation (Wang et al., 2017), thereby improving utility. After B_u is constructed, each bit is independently perturbed as:

$$\Pr[B'_u[i] = 1] = \begin{cases} \frac{1}{2}, & \text{if } B_u[i] = 1 \\ \frac{1}{e^{\epsilon} + 1}, & \text{if } B_u[i] = 0 \end{cases} \quad (13)$$

The perturbed bitvector B'_u is sent to the server. After collecting perturbed bitvectors from all users in $\mathcal{P}$, the server computes $\hat{C}(v)$ as in Eq. (11), which counts the total number of reported 1's at position v . The frequency estimate is then obtained as:

$$\hat{f}(v) = \frac{2((e^{\epsilon} + 1)\hat{C}(v) - |\mathcal{P}|)}{(e^{\epsilon} - 1)|\mathcal{P}|} \quad (14)$$

Subset Selection (SS): In the SS protocol, each user $u \in \mathcal{P}$ reports a subset $Z_u \subseteq D$ to the server. Let $s = |Z_u|$ denote the subset size parameter (default value is $s = \frac{|D|}{e^{\epsilon} + 1}$). User u initializes subset $Z_u = \emptyset$ and adds the true value v_u to Z_u with probability:

$$\sigma_s = \frac{s e^{\epsilon}}{s e^{\epsilon} + |D| - s} \quad (15)$$

The remaining elements of Z_u are sampled uniformly at random without replacement from $D \setminus \{v_u\}$ and added to Z_u . If v_u was included in Z_u , then $s - 1$ additional elements are sampled; otherwise, s elements are sampled. The user sends the resulting Z_u to the server.

After collecting all subsets Z_u from users u in $\mathcal{P}$, the server computes:

$$\text{Sup}(v) = |\{u \in \mathcal{P} : v \in Z_u\}| \quad (16)$$

The frequency estimate $\hat{f}(v)$ is given by:

$$\hat{f}(v) = \frac{\text{Sup}(v) - |\mathcal{P}|\theta_s}{(\sigma_s - \theta_s)|\mathcal{P}|}, \quad \text{where } \theta_s = \frac{(s - 1)s e^{\epsilon} + (|D| - s)s}{(|D| - 1)(s e^{\epsilon} + |D| - s)} \quad (17)$$

4.2. Post-processing module

Post-processing (PP) methods post-process the LDP frequency estimates $\hat{f}(v)$ and produce the post-processed frequencies $\tilde{f}(v)$ with the objective of improving statistical consistency and enhancing utility (Jia and Gong, 2019; Wang et al., 2020). Different PP strategies introduce different bias-variance trade-offs; consequently, their effectiveness may vary depending on the underlying dataset and the LDP protocol used to generate $\hat{f}(v)$. To support systematic evaluation and comparison of

PP methods, LDP³⁺ incorporates multiple widely adopted PP methods from the literature, which are described below. The Post-Processing Module is also designed to be extensible, allowing new PP methods to be added in the future.

Base-Pos: Frequencies must be non-negative by definition; however, due to the perturbation in LDP, $\hat{f}(v)$ may be negative for some $v \in D$. Base-Pos addresses this problem by converting all negative estimations to 0.

$$\forall v \in D : \tilde{f}(v) = \begin{cases} \hat{f}(v) & \text{if } \hat{f}(v) \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

Norm: The sum of frequencies across all $v \in D$ should equal 1; however, this may not hold due to the perturbation in LDP. Norm addresses this problem by adding a constant σ to each frequency so that the sum will equal 1.

$$\forall v \in D : \tilde{f}(v) = \hat{f}(v) + \sigma, \quad \text{such that } \sum_{v \in D} \tilde{f}(v) = 1 \quad (19)$$

Norm-Cut: Norm-Cut converts negative and small positive frequencies to 0, and it also ensures that the sum of frequencies equals 1. That is:

$$\forall v \in D : \tilde{f}(v) = \begin{cases} 0 & \text{if } \hat{f}(v) \leq \theta \\ \hat{f}(v) & \text{if } \hat{f}(v) > \theta \end{cases} \quad (20)$$

where θ is a threshold value. The value of θ is chosen such that $\sum_{v \in D} \tilde{f}(v) = 1$ is ensured.

Norm-Sub: Norm-Sub converts negative frequencies to 0. Then, it adds a constant δ to the frequencies to ensure that the sum of frequencies equals 1.

$$\forall v \in D : \tilde{f}(v) = \begin{cases} 0 & \text{if } \hat{f}(v) < 0 \\ \hat{f}(v) + \delta & \text{if } \hat{f}(v) \geq 0 \end{cases} \quad (21)$$

Here, the value of δ is chosen such that $\sum_{v \in D} \tilde{f}(v) = 1$ is ensured.

Norm-Mul: Norm-Mul converts negative frequencies to 0. Then, instead of an additive factor, it uses a multiplicative factor to the remaining frequencies so that their sum becomes 1.

$$\forall v \in D : \tilde{f}(v) = \begin{cases} 0 & \text{if } \hat{f}(v) < 0 \\ \alpha \hat{f}(v) & \text{if } \hat{f}(v) \geq 0 \end{cases} \quad (22)$$

Here, α is the multiplicative factor. Its value is chosen such that $\sum_{v \in D} \tilde{f}(v) = 1$ is ensured.

Power: The rationale of Power is that many real-world datasets follow a statistical distribution such as a Gaussian or power law distribution. Therefore, Power fits a distribution to the estimated frequencies and aims to minimize the expected square error. For example, with $\hat{f}(v) \sim P$ where P is a power law distribution, the goal is to minimize:

$$\min_P \mathbb{E} \left[\sum_{v \in D} (\hat{f}(v) - \tilde{f}(v))^2 \right] \quad (23)$$

Power-NS: Power-NS first applies Power and then uses Norm-Sub on Power's outputs to obtain the final result.

4.3. Adversarial module

In recent years, several poisoning and inference attacks against LDP protocols have been proposed in the literature Gursoy et al. (2022), Cheu et al. (2021), Cao et al. (2021), K. Huang et al. (2024) and Gürsoy (2024). These attacks have varying effectiveness depending on the underlying LDP protocol; furthermore, we observed that the use of PP methods can also have direct and indirect impacts on attack effectiveness. Motivated by these observations, we added the Adversarial Module to LDP³⁺, which is a key extension compared to our original LDP³ (Khodaie et al., 2025). The Adversarial Module of LDP³⁺ contains three representative and complementary classes of attacks that capture different attack dimensions: (i) poisoning attacks that aim to manipulate frequency estimates, (ii) budget inference attacks that attempt to

recover the privacy parameter ϵ , and (iii) longitudinal attacks that exploit repeated observations from the same user to infer private values. Below, we briefly describe these attacks.

Maximal Gain Attack (MGA) is a poisoning attack against LDP frequency estimation in which the attacker injects malicious users into the system (Cao et al., 2021). Recall that $\mathcal{P}$ denotes the population of honest users, and let $\mathcal{P}_{adv}$ denote the set of injected malicious users. The server aggregates reports from the combined population $\mathcal{P} \cup \mathcal{P}_{adv}$, and the reports sent from $\mathcal{P}_{adv}$ cause the poisoning effect. The attacker's objective in MGA is to maximize the estimated frequencies of a set of target items $T = \{\bar{v}_1, \bar{v}_2, \dots, \bar{v}_l\}$ where $T \subset \mathcal{D}$ by carefully creating the reports submitted by malicious users. In a default MGA where PP is not used, the attack's impact can be formalized as follows. Let $\hat{f}^p(v)$ denote the poisoned frequency estimate obtained from the combined population $\mathcal{P} \cup \mathcal{P}_{adv}$. The gain of the attack is defined as:

$$g^p(\bar{v}) = \hat{f}^p(\bar{v}) - f(\bar{v}) \quad (24)$$

Therefore, in MGA, the attacker's goal is to maximize $\sum_{\bar{v} \in T} g^p(\bar{v})$. Depending on the LDP protocol, the attacker can choose the strategy that maximizes this sum. On the other hand, when PP is used by the server, the end result is no longer $\hat{f}^p(\bar{v})$, but rather, it is $\tilde{f}^p(\bar{v})$ after post-processing. Consequently, the gain of the attack under PP needs to be defined as:

$$g^{pp}(\bar{v}) = \tilde{f}^p(\bar{v}) - f(\bar{v}) \quad (25)$$

Since $g^{pp}(\bar{v})$ is different from $g^p(\bar{v})$ and depends on the choice of PP method, it becomes interesting to measure the varying impacts of different PP methods on MGA.

Budget Inference Attack (BIA) aims to infer the privacy budget ϵ used by an LDP protocol (Baloglu and Gursoy, 2026). In the informed adversary setting, the attacker is assumed to know the true frequency distribution, collectively denoted by $\mathbf{f} = \{f(v)\}_{v \in \mathcal{D}}$, and the estimated distribution after LDP $\hat{\mathbf{f}} = \{\hat{f}(v)\}_{v \in \mathcal{D}}$. The attacker first computes a reference distance between the true and estimated distributions $\text{DIST}(\mathbf{f}, \hat{\mathbf{f}})$ using a chosen distance metric, e.g., ℓ_1 or ℓ_2 distance. To infer ϵ , the attacker performs a binary search-like process over an interval $[\epsilon_{min}, \epsilon_{max}]$. For each candidate ϵ_{cand} value in this range, the attacker generates a synthetic user population whose true distribution matches $\mathbf{f}$. Then, the attacker simulates the full data collection pipeline with the synthetic population: applying the LDP protocol with budget ϵ_{cand} , aggregating the reports, and performing estimation. This produces the simulated estimate $\hat{\mathbf{f}}_{cand}^{syn}$ with the distance $\text{DIST}(\mathbf{f}, \hat{\mathbf{f}}_{cand}^{syn})$. The goal of attacker's search is to find the ϵ_{cand} that yields highest similarity between $\text{DIST}(\mathbf{f}, \hat{\mathbf{f}}_{cand}^{syn})$ and $\text{DIST}(\mathbf{f}, \hat{\mathbf{f}})$.

The above process summarizes BIAs when PP is *not* used, as in the original paper Baloglu and Gursoy (2026). When PP is used, the attacker observes $\tilde{\mathbf{f}} = \{\tilde{f}(v)\}_{v \in \mathcal{D}}$ instead of $\hat{\mathbf{f}}$. Therefore, the above process must be performed with $\tilde{\mathbf{f}}$ in place of $\hat{\mathbf{f}}$, implying that the attack effectiveness will change according to which PP method is used, which makes the involvement and analysis of BIAs in LDP³⁺ interesting.

Longitudinal Attacks target iterative data collection settings where a user participates in multiple rounds of LDP data collection (Gursoy, 2024; Erlingsson et al., 2014; Arcolezi et al., 2022a). Unlike single-shot data collection, repeated observations of perturbed reports from the same user may enable inference of the user's true value. Consider a user $u \in \mathcal{P}$ whose true value is $v_u \in \mathcal{D}$. Let $\mathcal{O}_u = \{y_u^{(1)}, y_u^{(2)}, \dots, y_u^{(n)}\}$ denote the set of n perturbed reports observed from user u across multiple collection rounds. The attacker's goal is to infer v_u from $\mathcal{O}_u$. Following the Bayes rule, the prediction v_u^{pred} is computed as:

$$\begin{aligned} v_u^{pred} &= \underset{v \in \mathcal{D}}{\operatorname{argmax}} \Pr[v \mid \mathcal{O}_u] = \underset{v \in \mathcal{D}}{\operatorname{argmax}} \frac{\Pr[\mathcal{O}_u \mid v] \cdot \Pr[v]}{\Pr[\mathcal{O}_u]} \\ &= \underset{v \in \mathcal{D}}{\operatorname{argmax}} \Pr[\mathcal{O}_u \mid v] \cdot \Pr[v] \end{aligned} \quad (26)$$

The $\Pr[\mathcal{O}_u \mid v]$ term depends on which LDP protocol is used. The $\Pr[v]$ term can be found as $\Pr[v] = \frac{\tilde{f}(v)}{\sum_{v' \in \mathcal{D}} \tilde{f}(v')}$ when PP is not used, and

$\Pr[v] = \frac{\tilde{f}(v)}{\sum_{v' \in \mathcal{D}} \tilde{f}(v')}$ when PP is used. This implies that the attacker's predictions are also affected by whether PP is used, and if so, which PP method is used. Hence, studying longitudinal attacks in LDP³⁺ also becomes relevant and interesting.

4.4. Metrics module

The Metrics Module of LDP³⁺ provides quantitative metrics for evaluating the utility and adversarial effectiveness of LDP protocols and PP methods. Specifically, it consists of two categories of metrics: (i) utility metrics, which assess the discrepancy between true and estimated frequencies, and (ii) adversarial metrics, which quantify the success of the attacks. Below, we first describe the utility metrics implemented in LDP³⁺, including ℓ_1 distance, ℓ_2 distance, Kullback–Leibler divergence, and Earth Mover's Distance (EMD). These metrics are presented in terms of measuring the difference between the true frequencies $f(v)$ and the post-processed $\tilde{f}(v)$; however, they can trivially be modified and applied to measure the difference between $f(v)$ and $\hat{f}(v)$. Second, we describe the adversarial metrics for evaluating the effectiveness of attacks such as MGA, BIA, and longitudinal attacks.

Utility metrics: LDP³⁺ contains the implementations of five utility metrics, which are also commonly used in the literature. The first metric is ℓ_1 distance, also known as Manhattan distance, which is used to measure error using the absolute value norm. It is defined as:

$$\ell_1 \text{ distance} = \sum_{v \in \mathcal{D}} |\tilde{f}(v) - f(v)| \quad (27)$$

ℓ_2 distance, also known as Euclidean distance, measures error using the squared norm. It is defined as:

$$\ell_2 \text{ distance} = \sqrt{\sum_{v \in \mathcal{D}} (\tilde{f}(v) - f(v))^2} \quad (28)$$

Kullback–Leibler divergence is used to measure how much one probability distribution diverges from another. In our context, $f(v)$ and $\tilde{f}(v)$ are treated as probability mass functions, and KL-divergence is defined as:

$$\text{KL-divergence} = \sum_{v \in \mathcal{D}} f(v) \log \left(\frac{f(v)}{\tilde{f}(v)} \right) \quad (29)$$

Earth Mover's Distance (EMD) measures the minimum cost of transforming one distribution into another. It can be interpreted as the amount of work required to transform the original distribution $f(v)$ into the post-processed distribution $\tilde{f}(v)$. Mathematically, it is defined as:

$$\text{EMD} = \min_{\pi} \sum_{v_1, v_2 \in \mathcal{D}} \pi(v_1, v_2) \cdot |v_1 - v_2| \quad (30)$$

where $\pi(v_1, v_2)$ represents the amount of mass moved from v_1 to v_2 , and $|v_1 - v_2|$ is the distance between v_1 and v_2 . The distance $|v_1 - v_2|$ presumes that the domain $\mathcal{D}$ carries an order or a valid distance function, such as a numeric attribute domain (e.g., age, salary). For categorical domains without an inherent order, a domain ordering or distance function must be defined before EMD is applied.

Kendall τ is a metric for measuring rank correlation. In our context, it quantifies the agreement between the frequency ranking induced by the true frequencies $f(v)$ and the ranking induced by post-processed $\tilde{f}(v)$. For a pair of elements $v_i, v_j \in \mathcal{D}$, the pair is said to be:

$$(v_i, v_j) \text{ is } \begin{cases} \text{concordant} & \text{if } (f(v_i) - f(v_j))(\tilde{f}(v_i) - \tilde{f}(v_j)) > 0 \\ \text{tied} & \text{if } f(v_i) = f(v_j) \text{ and } \tilde{f}(v_i) = \tilde{f}(v_j) \\ \text{discordant} & \text{otherwise} \end{cases} \quad (31)$$

In particular, this means that a pair with $\tilde{f}(v_i) = \tilde{f}(v_j)$ while $f(v_i) \neq f(v_j)$ is counted as discordant, since it fails to reproduce the true relative order of v_i and v_j . Then, the Kendall τ metric is defined as:

$$\text{Kendall } \tau = \frac{(\# \text{ of concordant pairs}) - (\# \text{ of discordant pairs})}{|D| \cdot (|D| - 1)/2} \quad (32)$$

We note that treating such pairs as discordant is a conservative convention with respect to PP methods such as Norm-Cut, which map many low-frequency items to zero estimates and therefore produce many pairs with $\tilde{f}(v_i) = \tilde{f}(v_j)$. Under this convention, every such pair is penalized. As discussed in Section 5.3, Norm-Cut's rank preservation advantage persists despite this conservative treatment, indicating that it is not an artifact of this tie handling convention.

Adversarial metrics: LDP³+ contains the implementations of three adversarial metrics: *Gain Reduction (GR)* for MGAs, *Budget Inference Error (BIE)* for BIAs, and *Longitudinal Attack Success Rate (LASR)* for longitudinal attacks.

Gain Reduction (GR) quantifies the extent to which PP mitigates the impact of poisoning attacks. Recall that g^p denotes the gain of the attack without PP, and g^{pp} denotes the gain of the attack with PP. Given the target set $T \subseteq D$ of the attack, we define GR as:

$$GR = \sum_{\bar{v} \in T} (g^p(\bar{v}) - g^{pp}(\bar{v})) \quad (33)$$

In words, GR measures the reduction in the poisoned estimation gain on the target set that is achieved by post-processing. Since the true frequency $f(\bar{v})$ cancels in the difference above, GR can be written equivalently as $GR = \sum_{\bar{v} \in T} (\hat{f}^p(\bar{v}) - \hat{f}^{pp}(\bar{v}))$, i.e., the extent to which post-processing pulls the poisoned target estimates back. We interpret GR as an end-to-end measure of the defensive benefit of post-processing under attack rather than a fully isolated attack-specific metric, since post-processing may also modify estimates through its ordinary normalization behavior.

Budget Inference Error (BIE) measures the accuracy of BIAs by quantifying the difference between the true privacy budget used in data collection versus the budget inferred by the attacker. Let ϵ denote the true privacy budget and let ϵ^* denote the budget inferred by the attacker through BIA. We define:

$$BIE = |\epsilon^* - \epsilon| \quad (34)$$

A smaller BIE indicates a more successful attack, with $BIE = 0$ corresponding to perfect inference of the privacy budget.

Longitudinal Attack Success Rate (LASR) measures the success of longitudinal attacks in predicting the users' true values. It is defined as the ratio of the population for which the attacker's predicted v_u^p is correct:

$$LASR = \frac{\# \text{ of users } u \in \mathcal{P} \text{ such that } v_u = v_u^{pred}}{|\mathcal{P}|} \quad (35)$$

4.5. Multi-threading module

Identifying which LDP protocol and PP method combination yields the highest utility or lowest attack success rate requires the evaluation of numerous combinations, e.g., 6 protocols $\times$ 7 PP methods yields 42 possible combinations. Moreover, due to the inherent randomness of LDP, each combination must be executed multiple times to obtain statistically reliable results. Conducting such extensive experimentation in a purely serial manner can lead to high execution times. To mitigate this issue and ensure that each execution completes in a shorter time, LDP³+ supports multi-threading with the following workflow.

First, the dataset is partitioned into η equally sized partitions, where η corresponds to the number of available CPU threads. Each data partition is assigned to a separate thread. Second, within each thread, the user values in the assigned partition undergo LDP perturbation using the selected protocol (e.g., GRR, OLH, RAPPOR). Aggregation and estimation are performed independently inside each thread, i.e., the threads execute concurrently without interaction. Each thread runs with the same global parameters such as D , ϵ , and other protocol-specific parameters, guaranteeing consistent protocol behavior across parallel threads. This results in estimations $\hat{f}_1, \hat{f}_2, \dots, \hat{f}_\eta$, i.e., one estimation result per thread. Third, the per-thread estimates are averaged to obtain the population-wide frequency estimate $\hat{f}$. Considering that

the data partitions are of equal size, simply averaging the $\hat{f}_1, \hat{f}_2, \dots, \hat{f}_\eta$ to obtain $\hat{f}$ is sufficient. However, if partition sizes are different, then a weighted averaging proportional to partition size would ensure correctness. Finally, the resulting $\hat{f}$ is processed by the selected PP method(s) from the Post-Processing Module, producing the final post-processed frequencies $\tilde{f}$.

An important note is that our multi-threading is not limited to frequency estimation and post-processing; adversarial procedures are also parallelized. (This is also a distinguishing factor of LDP³+.) We observed that several aspects of MGA, BIA, and longitudinal attacks can be parallelized, and we took advantage of these observations. For example, in MGA, the set of malicious users is partitioned across threads, where each thread processes an equal portion of malicious users and optimizes those users' attacks. Since malicious report generation and perturbation are performed independently per user, this parallelism enables concurrent execution without loss of correctness. Similarly, the attacker's simulation process in BIAs (when computing $\hat{f}_{cand}^{syn}$) can be parallelized. In longitudinal attacks, since the prediction of v_u^{pred} depends only on the current user's $\mathcal{O}_u$, users can again be divided into separate threads. The $\Pr[v]$ term is population-wide; therefore, it needs to be computed once at the beginning, and all threads can inherit it.

4.6. Execution module

This module provides an interface for executing LDP³+ in practice. LDP³+ is operated through a single entry point (`main.py`) that exposes five subcommands: `estimate`, `poisoning`, `bia`, `rank`, and `longitudinal`, each corresponding to a different experimental task. Experiment parameters are managed through YAML-based configuration files, where task-specific configurations inherit shared defaults from a common base file. Any parameter can also be overridden directly via the command line. The general command structure and some examples in which the parameters are provided via the command line are shown in Fig. 2. As exemplified in Fig. 2, there exist several options and parameters:

- `--config` specifies the YAML configuration file for the selected task.
- `--epsilon` determines the privacy budget ϵ . One or more values can be provided; for tasks such as `poisoning` and `rank`, providing multiple values means a new experiment will run for each ϵ value.
- `--protocol` specifies the LDP protocol(s) to use, selected from those implemented in the Protocol Module. Specifying `'all'` runs experiments with all available protocols.
- `--method` specifies the PP method(s) to use, selected from those implemented in the Post-Processing Module. Specifying `'all'` runs experiments with all available PP methods.
- `--repeat` determines the number of repetitions per experiment to account for the inherent randomness of LDP and achieve statistical reliability.
- `--threads` specifies the number of parallel threads to use in multi-threaded execution.
- `--dataset` specifies the dataset path. Datasets are expected to contain one user value per row; the data reader can be modified to support alternative formats.
- `--metric` specifies the utility metric, selected from those implemented in the Metrics Module.
- `--num-malicious` specifies the number of malicious users to inject in the poisoning attack (`poisoning` subcommand). There also exist multiple pre-defined strategies for selecting the target set T from D . For example, `bottom_k` causes T to consist of the lowest-frequency items, and `top_k` causes T to consist of the highest-frequency items from D .
- `--run-bia` activates the binary search of the Budget Inference Attack (`bia` subcommand). The search bounds ϵ_{min} and ϵ_{max} should be provided through the configuration file.

```
python main.py <task> --config configs/<task>.yaml [options]

python main.py estimate --config configs/estimation.yaml \
  --dataset Datasets/Adult.csv --protocol olh grr \
  --method all --epsilon 0.5 1.0 2.0 --metric ll \
  --repeat 20 --threads 8

python main.py poisoning --config configs/mga.yaml \
  --dataset Datasets/Adult.csv --protocol olh \
  --method all --epsilon 0.5 1.0 2.0 3.0 4.0 \
  --num-malicious 1000 --repeat 20 --threads 8
```

Fig. 2. Sample commands to execute LDP³⁺. The generic, base command structure is given in the first command. A sample command to execute the frequency estimation task is given in the second command. (The task will be executed on the Adult dataset, using GRR and OLH as the protocols, all PP methods, $\epsilon = 0.5, 1.0$, and 2.0 , and ℓ_1 distance as the metric.) A sample command to execute the MGA attack (which is a poisoning attack) is given in the third command.

- `--num-observations` specifies the number of observed perturbed reports per user in the longitudinal attack (longitudinal subcommand).

5. Experiments and analysis

5.1. Experiment setup

In this section, we perform experiments with LDP³⁺ to analyze the varying utility and adversarial impacts of PP methods. Our experimental evaluation is organized around three main perspectives. First, we analyze utility under standard frequency estimation and rank preservation tasks, focusing on how the choice of LDP protocol, privacy budget ϵ , and PP method jointly shape performance. Second, we study sensitivity to experimental conditions by examining how data characteristics (e.g., distribution of the data and domain size $|D|$) and the selected utility metric influence which PP methods are preferred. Third, we evaluate robustness under adversarial settings by measuring how different PP methods transform the behavior and success of the implemented attacks (MGA, BIA, and longitudinal attacks) under adversarial metrics.

We used three real-world datasets (BMS-POS, Kosarak, Adult) and three synthetically generated datasets (Zipfian, Gaussian, Uniform) that are also commonly used in the DP and LDP literatures for experimentation.

Adult is a well-known dataset in the privacy literature, consisting of 45,222 records and 14 features. We downloaded it from the UCI ML Repository.² The features relate to individuals' census information such as age, education, workclass, marital status, race, and gender. In our experiments, we used *age* values as users' true values. Ages in the dataset ranged between 17 and 90; therefore, we have $|D| = 74$.

Kosarak contains click-stream data of a Hungarian online news portal. We downloaded it from the SPMF Dataset Repository.³ Each row in the dataset corresponds to one user's URL visit sequence. Due to many URLs having few occurrences (e.g., one or two), we pre-processed the dataset by identifying the top 128 most visited URLs and removed the rest, i.e., $|D| = 128$. For users who had more than one URL in their resulting stream, the most frequently occurring URL in their stream was picked as their v_u .

BMS-POS includes market basket sales data from a major electronics retailer, consisting of 515,596 transactions and 1657 unique items sold. We downloaded it from the public Github repository⁴ and pre-processed

it in a similar fashion to Kosarak by keeping only the top 256 most frequently purchased items.

Gaussian: We synthetically generated multiple Gaussian datasets in which users' true values are sampled from a Gaussian distribution with mean $\mu = 50$ and standard deviation $\sigma = 1, 5, 10$, and 20 . In all Gaussian datasets, the domain size is fixed to $|D| = 100$ and the population size is fixed to $|P| = 100,000$.

Zipfian: We generated Zipfian datasets in which users' true values are sampled from a Zipfian distribution. The Zipfian distribution is characterized by a skewness parameter s . In our datasets, we fixed $s = 1.5$ and $|P| = 1,000,000$, and varied the domain sizes $|D| = 32, 64, 128, 256, 512, 1024$, and 2048 .

Uniform: Finally, we generated Uniform datasets in which users' true value distribution follows a Uniform distribution. We set $|P| = 100,000$ by default and varied the domain size $|D|$.

5.2. Impacts of LDP protocols and privacy budgets

In Fig. 3, we present the ℓ_1 distance results of different PP methods under varying datasets, protocols, and ϵ budgets. Results in this figure are the average ℓ_1 distances over 20 runs. We observe from the plots in Fig. 3 that when ϵ is smaller (such as 0.5 or 1), there exist larger differences between the ℓ_1 distances of different PP methods. Furthermore, the benefits of using the best-performing PP method compared to not using PP (represented by the dashed line "No PP") or using a suboptimal PP method are substantial. (Throughout the paper, No PP stands for the LDP estimation results $f(v)$ without post-processing.) For example, on the Adult dataset, the best-performing PP method under $\epsilon = 0.5$ is Norm-Mul, and its ℓ_1 distance is almost half of No PP, Power, and Power-NS. However, as ϵ increases, we observe that the ℓ_1 distances of all PP methods converge, and they become similar to No PP. Considering that increased ϵ implies reduced perturbation, these results suggest that the need for PP methods is also reduced in the case of larger ϵ values. Since many PP methods converge as ϵ increases, achieving the best utility also becomes less dependent on the PP method used. Combining these results, we arrive at the following take-away message.

PP methods can yield substantial utility benefits, especially when ϵ is small. As ϵ becomes larger, PP methods' utility results converge to one another, and their benefits (over not using PP) decrease.

In Table 2, we provide results regarding which PP method performs best under varying datasets, ϵ budgets, and LDP protocols. The name of the best-performing PP method is written in each corresponding cell. Note that the cells are colored. We use darker colors to indicate how frequently, across 20 experiment runs, the best-performing PP method

² <https://archive.ics.uci.edu/dataset/2/adult>

³ <https://www.philippe-fournier-viger.com/spmf/index.php?link=datasets.php>

⁴ <https://github.com/cpearce/HARM/blob/master/datasets/BMS-POS.csv>

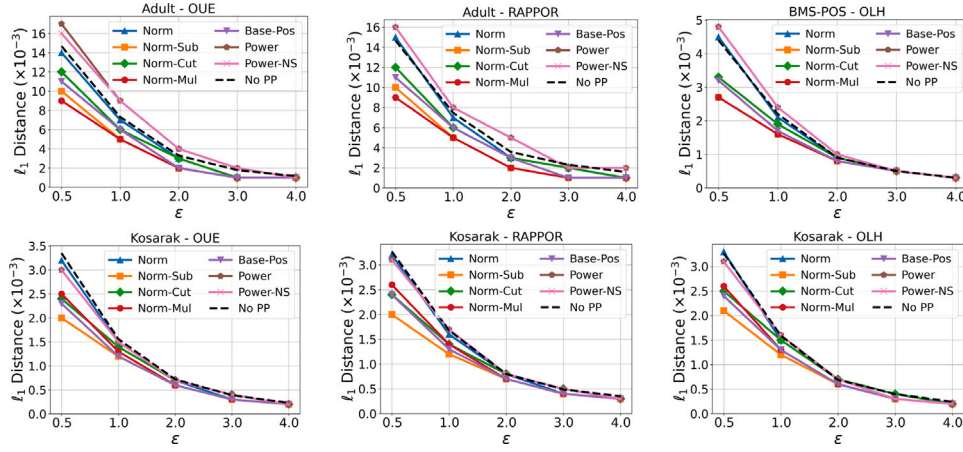


Fig. 3. Average utility loss (measured via ℓ_1 distance) of different PP methods under different datasets, protocols, and ϵ budgets. First column uses OUE protocol, second column uses RAPPOR protocol, and third column uses OLH protocol. Different datasets such as Adult, BMS-POS, and Kosarak are included for variety. ℓ_1 distances of the seven PP methods plus “No PP” (corresponds to LDP estimation results $\hat{f}(v)$ without post-processing) are each included as lines on the plots. X axis corresponds to different ϵ budgets, y axis corresponds to ℓ_1 distances.

Table 2

Best-performing PP method in terms of ℓ_1 distance under varying datasets (BMS-POS, Kosarak, Adult), LDP protocols (GRR, BLH, OLH, RAPPOR, OUE, SS), and ϵ budgets.

ϵ		GRR	BLH	OLH	RAPPOR	OUE	SS
$\epsilon = 0.5$	BMS-POS	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul
	Kosarak	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Power
	Adult	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul
$\epsilon = 1$	BMS-POS	Norm-Mul	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub
	Kosarak	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub
	Adult	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul
$\epsilon = 2$	BMS-POS	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Mul	Norm-Mul
	Kosarak	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub
	Adult	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Power
$\epsilon = 3$	BMS-POS	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Mul	Norm-Sub	Norm-Mul
	Kosarak	Norm-Sub	Norm-Sub	Base-Pos	Norm-Sub	Norm-Sub	Norm-Sub
	Adult	Power	Norm-Mul	Norm-Sub	Norm-Mul	Norm-Sub	Power
$\epsilon = 4$	BMS-POS	Norm-Mul	Norm-Sub	Base-Pos	Norm-Sub	Norm-Mul	Norm-Mul
	Kosarak	Base-Pos	Norm-Sub	Norm	Norm-Sub	Norm	Norm-Cut
	Adult	Power	Norm-Mul	Norm-Sub	Norm-Sub	Norm-Sub	Power

ends up being the best performer. For example, the cells corresponding to GRR and $\epsilon = 0.5$ are quite dark for all three datasets. This means that across 20 experiment runs, the PP methods written inside the cells (Norm-Mul and Norm-Sub) were the best performers in close to 20 of the runs. In contrast, the cells corresponding to OLH, RAPPOR, OUE, and SS with $\epsilon = 4$ are light-colored. This means that across 20 experiment runs, the PP methods written inside those cells were the best performers in fewer runs – although they are still the best, other PP methods perform closely.

We observe from Table 2 that Norm-Sub and Norm-Mul usually perform better than others, especially when $\epsilon = 0.5, 1$, and 2. Yet, they are not always the best. Power, Base-Pos, Norm, and others can be preferred under different settings. Thus, there is no single PP method that beats others in all settings. Furthermore, we observe that when ϵ is low (such as 0.5), the best PP method is relatively clear and consistent. Norm-Mul emerges as the best method for BMS-POS and Adult datasets, whereas Norm-Sub emerges as the best method for the Kosarak dataset. Considering the levels of darkness in the cells, we can conclude that both Norm-Mul and Norm-Sub are consistently the best performers. However, as ϵ gradually increases, two important trends emerge. First, the dominance of a single PP method starts to diminish. A single method does not outperform all others across different LDP protocols, i.e., different LDP protocols begin to favor different PP methods. This shift indicates that, as privacy levels increase, the preference of PP methods may become more protocol-dependent. Second, with larger values of ϵ , the frequency with which the best PP method produces the lowest ℓ_1 distance significantly decreases. It can be observed that

there are many more dark-colored cells when $\epsilon = 0.5$ compared to much lighter-colored cells when $\epsilon = 4$. This indicates that with larger values of ϵ , the effectiveness of PP methods becomes more variable, making it challenging to find a consistent PP method that performs better in a given setting.

We also highlight that the results obtained in Table 2 agree with the results in Fig. 3 in terms of which PP method performs best. Also, the convergence trend of different PP methods in Fig. 3 can help explain the lighter-colored cells in Table 2 when $\epsilon = 3$ or 4. In Fig. 3, we found that PP methods produce similar ℓ_1 distances when ϵ is large. Thus, the fact that the best-performing PP method changes frequently and the best-performing method is usually inconsistent in Table 2 can be explained by this finding from Fig. 3. Considering all results and discussion in this subsection, we can arrive at the following take-away message.

Although Norm-Sub and Norm-Mul typically perform well, the best-performing PP method can be different under different protocols, ϵ values, and datasets. The best-performing PP method is more consistent when ϵ is small, but becomes inconsistent as ϵ increases.

5.3. Impacts of PP on rank preservation

Next, we analyze whether PP methods provide benefits in the rank preservation task, and if so, which PP method is best to use. In Fig. 4, we present the Kendall τ results of various PP methods across different datasets, LDP protocols, and ϵ values. We observe that as ϵ increases,

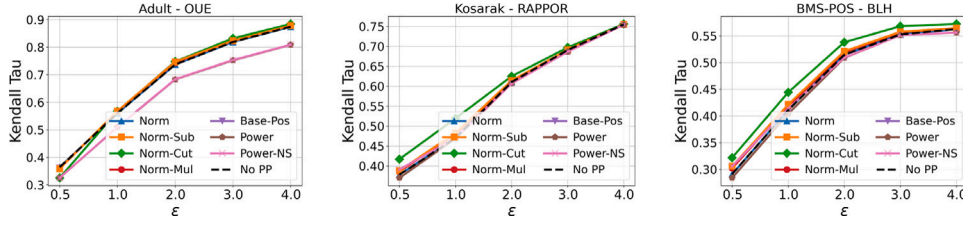


Fig. 4. Average rank correlation (measured via Kendall τ) of different PP methods under various datasets and ϵ budgets. Higher values indicate better preservation of item rankings.

Table 3

Best-performing PP method in terms of Kendall τ rank correlation under varying datasets (BMS-POS, Kosarak, Adult), LDP protocols (GRR, BLH, OLH, RAPPOR, OUE, SS), and ϵ budgets.

ϵ		GRR	BLH	OLH	RAPPOR	OUE	SS
$\epsilon = 0.5$	BMS-POS	Norm	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Kosarak	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Adult	Norm-Cut	Norm	Norm-Cut	Norm	Base-Pos	Norm-Cut
$\epsilon = 1$	BMS-POS	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Kosarak	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Adult	Base-Pos	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
$\epsilon = 2$	BMS-POS	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Kosarak	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Adult	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
$\epsilon = 3$	BMS-POS	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Kosarak	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
	Adult	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut
$\epsilon = 4$	BMS-POS	Norm-Cut	Norm-Cut	Power	Norm-Cut	Power	Power
	Kosarak	Power	Norm-Cut	Power	Norm-Cut	Power	Power
	Adult	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Cut

the Kendall τ values for all methods improve, indicating that reduced perturbation naturally leads to better preservation of the relative item rankings. Norm-Cut generally maintains a lead over the No PP baseline as well as the other PP methods, which especially struggle to maintain high rank correlation in lower ϵ regimes. For instance, on the Kosarak and BMS-POS datasets, Norm-Cut significantly outperforms the other PP methods. On the Adult dataset, Power and Power-NS consistently fall below the No PP baseline. While most methods' rank preservation capabilities improve with higher budgets, some PP methods are much more reliable for rank preservation when privacy budgets are strict.

Combining these observations, we arrive at the following take-away message.

PP methods, particularly Norm-Cut, usually improve rank preservation compared to raw LDP estimates. This benefit is most visible for low to mid-range ϵ values.

In Table 3, we provide a detailed breakdown of the best-performing PP method for rank preservation under varying experimental settings. Similar to our frequency estimation analysis, the cells are colored to indicate the frequency with which a specific PP method emerged as the winner across 20 runs, with darker colors representing higher consistency.

The most notable result in Table 3 is the dominance of Norm-Cut. Across the BMS-POS and Kosarak datasets, Norm-Cut is almost exclusively the top performer for ϵ values between 0.5 and 3.0. The high density of dark-colored cells in these regions indicates that Norm-Cut is not just the best on average, but is consistently the superior choice in nearly every individual run. However, as we move to $\epsilon = 4$, we observe a transition similar to the one noted in the frequency estimation analysis. The dominance of Norm-Cut begins to diminish, and methods such as Power and Base-Pos start to emerge as the best performers, particularly for the BMS-POS and Adult datasets. Furthermore, the Adult dataset exhibits more variability at $\epsilon = 0.5$, where Norm-Cut, Norm, and Base-Pos share the top spots across different protocols. This suggests that while Norm-Cut is generally a good choice

for ranking, the optimal PP method can be sensitive to the specific dataset characteristics and the LDP protocol used. Based on these results, we can summarize our findings with the following take-away message.

Norm-Cut is the most effective PP method for rank preservation across most settings. However, at higher ϵ budgets, its dominance fades as other methods like Power become competitive or superior.

The reason Norm-Cut is well-suited to rank preservation is that rank errors are often caused by the low-frequency tail items rather than the high-frequency items. Since the LDP noise is roughly constant in absolute scale regardless of an item's frequency, in the tail it becomes comparable to the gaps between neighboring items, so many pairwise comparisons there are likely to be affected by noise. By zeroing every estimate at or below threshold θ (Eq. (20)), Norm-Cut collapses this noisy tail into ties. Consequently, it is not possible that a tail item beats a high-frequency item. It is also important to note that Norm-Cut remains superior to other PP methods despite its zeroing. Recall that, under our Kendall τ definition, a tied pair is counted as discordant, so pairs between two tail items that are tied by Norm-Cut but differ in their true frequencies decrease τ . The fact that Norm-Cut's superiority persists despite this shows that its suitability is not an artifact of how ties are handled in the Kendall τ definition.

5.4. Impacts of data characteristics

In this section, we analyze the impacts of two data-related characteristics: data distribution and domain size. To measure the impacts of the data distribution, we utilize the Gaussian and Uniform datasets. We generate Gaussian datasets with varying standard deviations ($\sigma = 1, 5, 10$, and 20) to experiment with varying degrees of non-uniformity in $f(v)$, and a Uniform dataset which in which all $f(v)$ are equal. Smaller σ means the data distribution is more concentrated around the mean, and therefore there is higher non-uniformity.

Table 4
Best-performing PP method under varying Gaussian and Uniform datasets.

	Protocols	Standard Deviation				
		$\sigma = 1$	$\sigma = 5$	$\sigma = 10$	$\sigma = 20$	Uniform
$\epsilon = 1$	GRR	Norm-Cut	Norm-Cut	Norm-Mul	Norm-Mul	Norm-Mul
	BLH	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Mul	Norm-Mul
	OLH	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Mul	Norm-Mul
	RAPPOR	Norm-Cut	Norm-Cut	Norm-Cut	Norm-Mul	Norm-Mul
	OUE	Norm-Cut	Norm-Cut	Norm-Sub	Norm-Mul	Norm-Mul
	SS	Norm-Cut	Norm-Cut	Norm-Sub	Norm-Mul	Norm-Mul

Table 5
Best-performing PP method under Zipfian datasets with varying $|D|$.

	Protocols	Domain Size $ D $						
		32	64	128	256	512	1024	2048
$\epsilon = 1$	GRR	Base-Pos	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Cut	Norm-Cut	Norm-Cut
	BLH	Norm	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Cut
	OLH	Norm	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Cut	Norm-Cut
	RAPPOR	Norm	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Cut	Norm-Cut
	OUE	Norm	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Sub	Norm-Cut	Norm-Cut
	SS	Norm-Cut	Norm-Mul	Norm-Sub	Power	Norm-Sub	Norm-Cut	Norm-Cut

The results of this experiment are shown in Table 4. We use the same representation and color coding as in Table 2. The ϵ value is 1. We observe from Table 4 that Norm-Cut is consistently the best-performing PP method for smaller σ values such as $\sigma = 1$ and 5. However, as σ increases to 10 and 20, PP methods such as Norm-Mul and Norm-Sub start becoming more preferable. For $\sigma = 20$ and the Uniform dataset, Norm-Mul becomes the best choice consistently. This progression reveals a clear trend: as non-uniformity changes, the best-performing PP method changes. Norm-Cut is the better choice when the data is non-uniform, whereas Norm-Mul becomes the better choice when the data is more uniform. This observation can be explained via the mathematical definitions of the PP methods as follows. When the data is uniform, $f(v)$ of all $v \in D$ are similar and there do not exist values with substantially lower frequencies than others. In this case, the multiplicative correction applied by Norm-Mul is sufficient to ensure consistency and reduce error. On the other hand, when data is non-uniform, there exist some values v such that $f(v)$ is 0 or close to 0. After LDP, some of these values end up with non-zero but low $\hat{f}(v)$. Norm-Cut applies the rule $\tilde{f}(v) = 0$ if $\hat{f}(v) \leq \theta$ to them, i.e., these values' frequencies are post-processed to 0. Thus, it reduces error especially in datasets with high non-uniformity.

In our second experiment, we analyze the impact of domain size $|D|$. For this experiment, we generate Zipfian datasets with varying $|D| = 32, 64, 128, 256, 512, 1024$, and 2048. We conduct a similar analysis to Tables 2 and 4. The results are shown in Table 5. For the smallest domain size $|D| = 32$, different protocols may favor different PP methods and the cells are not dark-colored, suggesting that a single PP method does not consistently outperform the others. As $|D|$ increases to 64, 128, and 256, Norm-Sub becomes more dominant for most protocols. When $|D|$ is further increased to 1024 and 2048, Norm-Cut emerges as the best-performing method. For the largest domain size $|D| = 2048$, Norm-Cut becomes the best-performing protocol consistently. The superior performance of Norm-Sub and Norm-Cut for larger $|D|$ agrees with results from Table 4, since both Norm-Sub and Norm-Cut post-process negative frequencies and low frequencies to 0. When $|D|$ is large, as $|P|$ is constant, there is a higher likelihood of obtaining low-frequency values $v \in D$, whose frequencies may remain low or become negative after LDP perturbation. Norm-Sub and Norm-Cut help in reducing the errors in these cases. Overall, the results also demonstrate that the best PP method changes when $|D|$ is changed, e.g., multiple PP methods perform well for small domain sizes, Norm-Sub emerges as the best approach for medium-sized domains (128 to 512), and Norm-Cut is the best for large domains (> 1024). Considering

Table 6
Best-performing PP method under varying utility metrics. Kosarak dataset and $\epsilon = 1$ are used.

	Protocols	Utility Metric		
		ℓ_1 distance	ℓ_2 distance	KL-divergence
$\epsilon = 1$	GRR	Norm-Sub	Norm-Sub	Norm-Sub
	BLH	Norm-Sub	Norm-Sub	Norm-Sub
	OLH	Norm-Sub	Norm-Sub	Norm-Sub
	RAPPOR	Norm-Sub	Norm-Sub	Norm-Sub
	OUE	Norm-Sub	Norm-Sub	Norm-Sub
	SS	Norm-Sub	Norm-Sub	Power

the results and discussion in this subsection, we arrive at the following take-away message.

The best-performing PP method is dependent on data characteristics, e.g., data distribution (non-uniformity) and domain size.

5.5. Impacts of utility metrics

In most of the utility experiments reported so far, ℓ_1 distance was used as the default utility metric. In this section, we examine how the choice of utility metric impacts the performance of PP methods. We use three utility metrics: ℓ_1 distance, ℓ_2 distance, and KL-divergence. We report experiments with varying protocols, fixed $\epsilon = 1$, and the Kosarak dataset. The results are given in Table 6. The same format and color coding as the previous tables are used.

For most protocols, ℓ_1 distance and ℓ_2 distance metrics consistently favor Norm-Sub. This observation underscores that the ℓ_1 and ℓ_2 distance metrics, which are similar in nature, agree in terms of the favorable PP method. In contrast, KL-divergence exhibits slightly more variability compared to ℓ_1 and ℓ_2 . While Norm-Sub still performs well for most protocols, there are notable deviations. For example, for SS, Power is consistently the best-performing PP method. The reduced darkness of the cells corresponding to KL-divergence further highlights the varying nature of KL-divergence in comparison to ℓ_1 and ℓ_2 distance. Overall, the results indicate that the best-performing PP method is indeed influenced by the choice of utility metric. Metrics that are inherently similar to one another tend to prefer the same PP methods, whereas different metrics can prefer different PP methods.

Norm-Sub typically performs well across multiple utility metrics, but the optimal PP method can change according to the choice of metric and protocol.

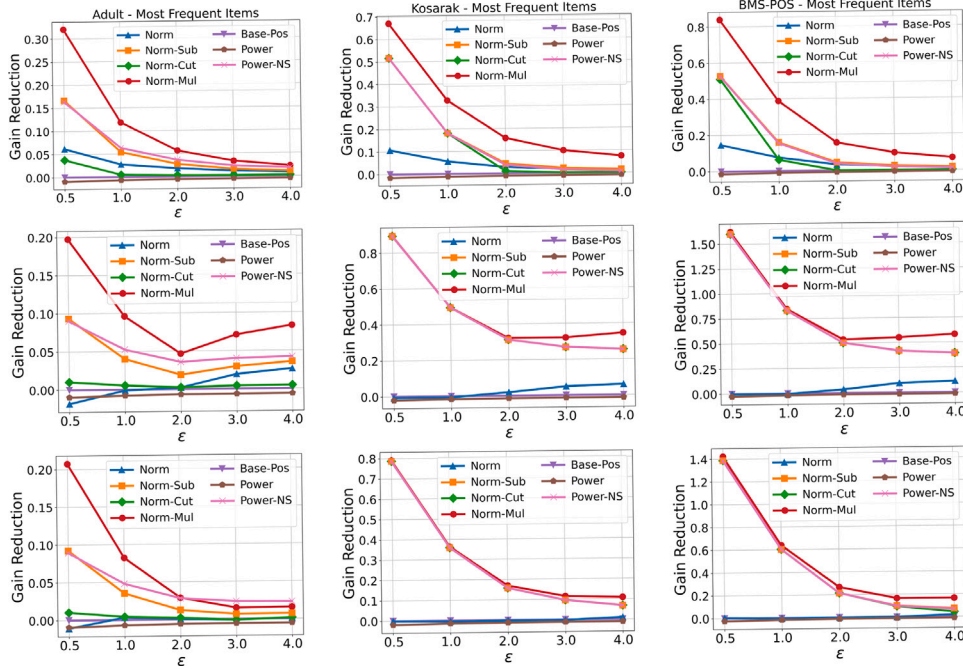


Fig. 5. Gain Reduction (GR) achieved by different PP methods under MGA. Results are given across three datasets (Adult, Kosarak, BMS-POS) and three protocols (top row: OLH; second row: OUE; bottom row: RAPPOR).

5.6. Impacts of PP on maximal gain attacks

Next, we evaluate the impacts of various PP methods on the Maximal Gain Attack (MGA) using the Gain Reduction (GR) metric. We set the cardinality of poisoning users as $|\mathcal{P}_{adv}| = 1\% \times |\mathcal{P}|$, and assume that the target set T is either the k most frequent items in $\mathcal{D}$ (top- k) or the k least frequent items in $\mathcal{D}$ (bottom- k). The value of k is set to 10, 20, and 40 for the Adult, Kosarak, and BMS-POS datasets respectively, due to their different $|\mathcal{D}|$. To align the adversarial evaluation with the breadth of the utility study, we report MGA results across three representative protocols: OLH, OUE, and RAPPOR. Results of experiments performed under varying ϵ , datasets, and PP methods are shown in Fig. 5.

Recall that a higher GR value indicates that the PP method more effectively suppresses the attacker's ability to inflate the frequencies of the target items, making GR a measure of “defense strength” against MGA. Across all three datasets and both target sets, GR values are highest when ϵ is small and decrease consistently and converge as ϵ increases. This mirrors the general utility trend observed in Fig. 3: as the privacy budget increases, LDP perturbation weakens, frequency estimates become more accurate, and PP methods have less noise to deal with, reducing both their utility benefit and their GR effect. Among the PP methods, Norm-Mul stands out most clearly, achieving GR values as high as 0.84 on BMS-POS and 0.67 on Kosarak at $\epsilon = 0.5$ when the most frequent items are targeted. Norm-Sub and Power-NS follow as the next strongest defenders, with Norm-Cut also performing well in several cases. On the other hand, GRs of Base-Pos and Power remain near zero throughout, indicating that they provide no reduction in the attacker's gain regardless of ϵ , dataset, or the target item set.

The near-zero GR of Base-Pos and Power warrants an explanation, particularly for Power, whose distributional fitting might be expected to help. The common underlying reason is that both methods process each item's estimate in isolation: the post-processed frequency of a target item depends only on that item's own estimate, and neither method moves mass between items. A successful MGA positively inflates target items' frequency estimates, and once an estimate is positive and inflated, a method that treats each item independently cannot pull it back down. Base-Pos only resets negative estimates to zero, so it reduces the attacker's gain only when a target estimate happens to

be negative (which is avoided by a successful MGA). Power fits a power-law distribution and adjusts each estimate accordingly, but it still adjusts each item on its own, based only on that item's frequency and the fitted distribution. Furthermore, since MGA concentrates the injected mass on a small set of target items, it barely changes the overall estimates from which the distribution is fitted, so the fitted distribution is not significantly changed and Power's impact is less significant. In contrast, the normalization-based PP methods (e.g., Norm-Mul) reduce the attacker's gain because enforcing the sum-to-one constraint ties all items together: lowering the inflated target estimates is forced by the need to rescale the sum of the whole estimation back to a total of one. Base-Pos and Power lack any such linkage between items, which is why they offer less protection. This also explains their slightly negative GR at low ϵ (e.g., on the Adult dataset): without a normalization constraint, they cannot remove the inflated mass and can only marginally suppress small positive noise on non-target items, which works against the GR goal. Finally, we note that Base-Pos and Power denoise and adjust estimates in both poisoned and/or non-poisoned settings, so their near-zero GR here is an indication that the GR metric captures the end-to-end reduction in poisoned estimation gain involving both the denoising effect of PP methods and the attack-related impacts.

In addition, comparing the three protocols (the three rows of Fig. 5), we observe that the impacts of PP methods under MGA are largely consistent across protocols. Norm-Mul, Norm-Sub, and Power-NS again emerge as the strongest defenders for OUE and RAPPOR, just as for OLH, achieving the highest GRs. For the unary encoding protocols these three methods perform very similarly, with their curves nearly overlapping at low ϵ . Base-Pos and Power again yield essentially no GR across all three protocols. The overall ϵ trend is also preserved, with GR highest at small ϵ , although OUE exhibits a less monotonic decline than OLH and RAPPOR. Upon analyzing the standard deviations of the GRs across 20 repetitions, we also observed that these trends are consistent, with low standard deviations. For example, the majority of the standard deviations for the Adult dataset results are lower than 0.01. Considering the clear ϵ and PP method trends observed in Fig.

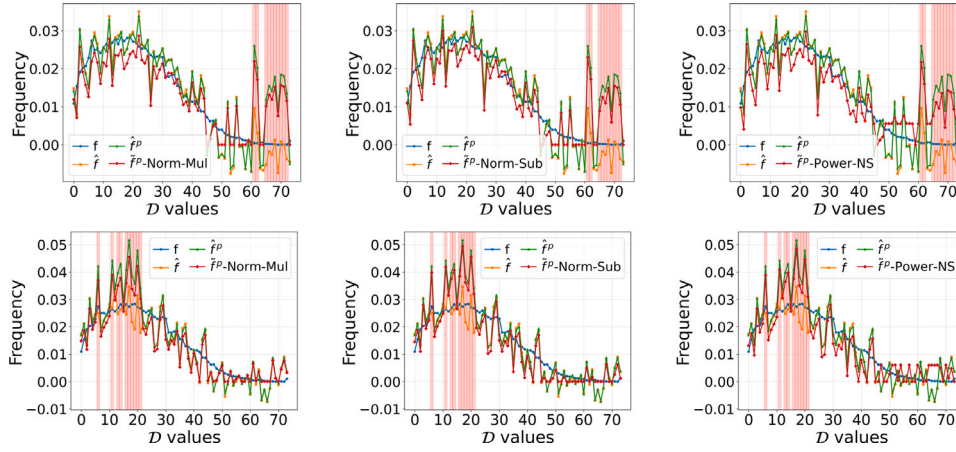


Fig. 6. Frequency distributions under MGA on the Adult dataset at $\epsilon = 1.5$ for Norm-Mul, Norm-Sub, and Power-NS (top row: top- k items targeted; bottom row: bottom- k items targeted). Red-shaded areas indicate the items targeted by the attacker.

5, such small variability is unlikely to change the main take-aways (e.g., whether a PP method achieves high or low GRs).

Different PP methods yield substantially different levels of gain reduction under MGA. Norm-Mul, Norm-Sub, and Power-NS (especially Norm-Mul) achieve the highest GRs across different datasets and target sets, especially at low ϵ .

To further understand how PP methods reduce the effect of MGA at the frequency level, Fig. 6 shows the estimated frequency distributions under attack for three best-performing PP methods (Norm-Mul, Norm-Sub, and Power-NS) on the Adult dataset at $\epsilon = 1.5$. Each plot overlays four curves: the true frequency f , the clean LDP estimate $\hat{f}$, the poisoned but non-post-processed estimate $\hat{f}^p$, and the post-processed poisoned estimate $\tilde{f}^p$. The red-shaded vertical bands mark the items targeted by the attacker (i.e., the items in T). In the top- k plots (top row), $\hat{f}^p$ visibly spikes above the true f at the shaded positions, confirming that the attacker successfully inflates those frequencies before post-processing. After PP is applied, $\tilde{f}^p$ is pulled back towards f at the shaded positions, with Norm-Mul, Norm-Sub, and Power-NS all achieving a noticeable correction. In the bottom- k plots (least frequent items), the targeted items sit in the tail of the distribution where frequencies are inherently small, and the poisoned frequencies again depart from f at the shaded positions. PP similarly draws $\tilde{f}^p$ closer to f in this region, though the absolute correction is smaller given that the starting frequencies were low. Across both target settings, the three methods behave similarly in their corrective effect, consistent with the GR results in Fig. 5, which placed them in close proximity at ϵ values near 1.5. In summary, our key take-away message from these experiments is as follows.

Post-processing is beneficial to defend against MGAs by pulling the poisoned frequency estimates towards the true distribution. This corrective effect is most pronounced at low ϵ where PP methods modify the estimates more aggressively.

Since f , $\hat{f}$, $\hat{f}^p$, and $\tilde{f}^p$ are all plotted in Fig. 6, this figure also enables us to compare clean versus adjusted frequencies in the corresponding non-poisoned setting (non-poisoned setting is captured by $\hat{f}$). For non-poisoned items, their clean estimated frequencies $\hat{f}$ can be an overestimation or underestimation of their true frequency f , due to randomness in LDP. This is also true for their $\hat{f}^p$. On the other hand, for poisoned items (i.e., items that are targeted by MGA), their poisoned $\hat{f}^p$ are significantly larger than their $\hat{f}$ and f due to the attack. The impact of PP on poisoned items and non-poisoned items can

therefore be different. For poisoned items, PP methods almost always reduce the inflated frequencies, i.e., their $\tilde{f}^p$ are lower than $\hat{f}^p$. In contrast, for non-poisoned items, PP methods may increase or decrease the frequencies. Notably, despite the positive GR of PP methods for poisoned items, $\tilde{f}^p$ values often still remain significantly higher than f and $\hat{f}$. This shows that PP methods help but do not fully mitigate the poisoning attack.

5.7. Impacts of protocols and PP methods on budget inference attacks

We also evaluate the vulnerability of LDP protocols and PP methods against BIAs using the BIE metric. Recall that a higher BIE indicates that the attacker's inferred budget deviates further from the true privacy budget, and therefore corresponds to a weaker attack. We report the BIE results for the Adult and Kosarak datasets in Tables 7 and 8, covering all six LDP protocols, seven PP methods, and five ϵ values. All BIE values reported in Tables 7 and 8 are averaged over 20 independent repetitions. For each protocol and ϵ combination, the PP method achieving the highest BIE is highlighted. Avg-PP denotes the average BIEs of all PP methods, and No-PP denotes the case without post-processing.

When we compare LDP protocols with one another, we observe that GRR consistently yields the lowest average BIEs across many ϵ levels and both datasets, meaning the attacker can infer the true privacy budget most accurately when GRR is used. BLH, on the other hand, exhibits substantially higher average BIE values that grow considerably with ϵ , reaching a column average of 0.296 at $\epsilon = 4.0$ on the Adult dataset and 0.359 on Kosarak. This behavior can be attributed to the binary hash encoding of BLH, which introduces hash collisions and randomness that are difficult for the attacker to simulate and/or overcome. For OLH, OUE, RAPPOR, and SS, we observe that their BIEs fall between the two ends (GRR and BLH). These results indicate that the choice of LDP protocol is significant in remaining robust against BIAs.

The choice of LDP protocol affects the effectiveness of BIAs. GRR has the highest vulnerability to BIAs, whereas BLH has the lowest vulnerability.

Next, we study the impacts of PP methods. Our first observation is that PP is usually (but not always) beneficial to achieve protection against BIAs. PP can increase the average BIE relative to No-PP, indicating that it may introduce additional uncertainty into the attacker's budget inference process. The margin of improvement can sometimes be substantial, e.g., under RAPPOR on the Adult dataset, the best PP method (Norm-Mul, 0.100) achieves nearly four times the BIE of No-PP

Table 7

BIE results on the Adult dataset for different LDP protocols and PP methods. For each protocol and ϵ , the highest BIE (most robust configuration) is highlighted. The PP method with the highest average BIE across all ϵ is in bold and underlined.

GRR						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.008	0.025	0.002	0.081	0.058	0.035
Norm	0.003	0.026	0.030	0.063	0.023	0.029
Norm-Cut	0.023	0.005	0.014	0.022	0.056	0.024
Norm-Mul	0.030	0.057	0.023	0.048	0.021	0.036
Norm-Sub	0.037	0.045	0.026	0.038	0.009	0.031
Power	0.001	0.056	0.005	0.016	0.055	0.027
Power-NS	0.010	0.005	0.029	0.070	0.008	0.024
Avg-PP	0.016	0.031	0.018	0.048	0.032	
No-PP	0.011	0.012	0.001	0.072	0.042	0.027
RAPPOR						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.002	0.061	0.057	0.034	0.098	0.050
Norm	0.026	0.009	0.061	0.038	0.109	0.049
Norm-Cut	0.032	0.001	0.094	0.041	0.236	0.081
Norm-Mul	0.055	0.037	0.145	0.077	0.187	0.100
Norm-Sub	0.009	0.056	0.073	0.056	0.213	0.081
Power	0.034	0.043	0.051	0.060	0.080	0.054
Power-NS	0.015	0.050	0.021	0.028	0.054	0.034
Avg-PP	0.025	0.037	0.072	0.048	0.140	
No-PP	0.001	0.029	0.022	0.041	0.047	0.028
BLH						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.021	0.008	0.008	0.232	0.101	0.074
Norm	0.019	0.004	0.133	0.009	0.349	0.103
Norm-Cut	0.022	0.010	0.168	0.021	0.500	0.144
Norm-Mul	0.069	0.033	0.280	0.065	0.242	0.138
Norm-Sub	0.002	0.018	0.121	0.043	0.242	0.085
Power	0.013	0.020	0.135	0.117	0.205	0.098
Power-NS	0.013	0.003	0.085	0.028	0.431	0.112
Avg-PP	0.023	0.014	0.133	0.074	0.296	
No-PP	0.001	0.055	0.104	0.165	0.233	0.111
OLH						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.015	0.024	0.065	0.004	0.037	0.029
Norm	0.001	0.072	0.001	0.108	0.051	0.047
Norm-Cut	0.016	0.015	0.107	0.034	0.019	0.038
Norm-Mul	0.017	0.065	0.070	0.012	0.044	0.042
Norm-Sub	0.037	0.042	0.099	0.151	0.012	0.068
Power	0.010	0.006	0.107	0.074	0.040	0.047
Power-NS	0.043	0.062	0.098	0.125	0.012	0.068
Avg-PP	0.020	0.041	0.078	0.073	0.031	
No-PP	0.013	0.009	0.077	0.037	0.051	0.037
OUE						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.010	0.031	0.077	0.042	0.166	0.065
Norm	0.021	0.049	0.068	0.076	0.047	0.052
Norm-Cut	0.046	0.026	0.009	0.002	0.088	0.034
Norm-Mul	0.004	0.052	0.064	0.050	0.091	0.052
Norm-Sub	0.013	0.040	0.061	0.025	0.147	0.057
Power	0.024	0.085	0.052	0.069	0.011	0.048
Power-NS	0.034	0.024	0.048	0.115	0.118	0.068
Avg-PP	0.022	0.044	0.054	0.054	0.095	
No-PP	0.021	0.052	0.068	0.022	0.045	0.041
SS						
Method	ϵ	0.5	1.0	2.0	3.0	4.0
Base-Pos	0.022	0.025	0.047	0.042	0.064	0.040
Norm	0.005	0.004	0.057	0.016	0.007	0.018
Norm-Cut	0.004	0.016	0.062	0.107	0.044	0.047
Norm-Mul	0.037	0.035	0.083	0.010	0.075	0.048
Norm-Sub	0.025	0.004	0.097	0.002	0.080	0.042
Power	0.019	0.001	0.010	0.078	0.075	0.037
Power-NS	0.026	0.072	0.008	0.019	0.061	0.037
Avg-PP	0.020	0.022	0.052	0.039	0.058	
No-PP	0.009	0.032	0.089	0.071	0.020	0.044

(0.028), and a similarly large gap is observed under BLH on Kosarak, where Norm-Cut (0.259) far exceeds No-PP (0.061). However, this benefit is not universal across all PP methods and settings, as several protocol-dataset combinations in Tables 7 and 8 show that Avg-PP does not exceed No-PP. Therefore, rather than blindly recommending the use of PP, we focus on which PP methods more reliably improve robustness, as detailed below.

Regarding which PP method is most effective, the results can be dataset-dependent. No single PP method consistently maximizes or minimizes BIEs across protocols and datasets. The PP method that yields the highest and lowest BIEs frequently differs from protocol to protocol and ϵ to ϵ , which can be observed from the highlighted cells frequently shifting across rows and columns in Tables 7 and 8. On the Adult dataset, Base-Pos and Norm-Mul are slightly better than the others, while on the Kosarak dataset, Norm-Cut appears to be the better choice. However, these trends also change according to the protocol, e.g., on the Adult dataset under GRR, Power-NS and Norm-Cut yield the lowest average BIEs (0.024 each), while under OUE the lowest BIEs belong to Norm-Cut (0.034), and under RAPPOR lowest BIEs are caused by Power-NS (0.034). On Kosarak, Power yields the lowest BIEs and Base-Pos yields the highest BIEs under GRR, though many PP methods' BIEs are similar.

There are several reasons behind this variability. Recall that the BIA infers ϵ by matching the distance between the true and observed distributions against the distances produced by simulated data collection at candidate ϵ values. PP may flatten this distance-versus- ϵ relationship or make it non-monotonic. Furthermore, each PP method distorts the distance-versus- ϵ relationship in a different way because the PP method's distortion also depends on the shape of the estimated distribution, which in turn depends on the combination of protocol, dataset, and ϵ . Consequently, any single method's distortion cannot be maximal across all of these combinations. As a result, the PP method that yields the highest BIE shifts from one setting to another, which is why the highlighted cells move across different rows and columns. We also acknowledge that BIAs adopted from the state-of-the-art work (Baloglu and Gursoy, 2026) are not guaranteed to behave optimally when PP makes the distance-versus- ϵ relationship non-monotonic. Thus, the reported BIEs should be interpreted as measuring the effectiveness of a current state-of-the-art BIA method (with its published search procedure), rather than an optimal search procedure or all possible search procedures. Nevertheless, the different behaviors of PP methods (i.e., their non-deterministic and non-monotonic distortion) would remain as a relevant reason why BIEs change, even if BIAs had a different

search strategy or a hypothetically optimal search strategy, since the $\text{DIST}(\mathbf{f}, \tilde{\mathbf{f}})$ term used in BIA's search would still differ from $\text{DIST}(\mathbf{f}, \hat{\mathbf{f}})$ and from one PP method to another.

Although the single best method is setting-dependent, a clearer and more actionable picture emerges when we ask which methods reliably outperform the No-PP baseline. Examining the average BIE columns across all protocol-dataset combinations in Tables 7 and 8, Norm-Mul exceeds No-PP in 11 of 12 cases and Norm-Cut in 10 of 12, with Norm-Cut additionally providing the largest average margin over No-PP. In contrast, methods such as Norm and Base-Pos exceed No-PP in only 7 of 12 cases and by negligible margins, making them less reliable. Hence, while no method is universally optimal, Norm-Mul and Norm-Cut stand out as the most consistent choices for improving robustness against BIAs. This is also consistent with our other findings, as Norm-Mul is the strongest method against MGA (Section 5.6) and Norm-Cut is the most effective for rank preservation (Section 5.3).

We also studied the standard deviations of the BIEs across 20 repetitions. While we do not report the numerical results in Tables 7 and 8 for brevity, we briefly discuss the main findings of our analysis. First, we found that the standard deviations are non-negligible. Consequently, the small BIE differences between individual PP methods within a single setting can be caused by run-to-run variability (i.e., randomness). Therefore, basing our take-aways on the aggregate trends across all 12 protocol-dataset combinations, rather than on individual cells, is important and justified. Second, although the standard deviations can be non-negligible, they oftentimes remain significantly below the true ϵ value, especially as ϵ increases. The main outlier is the BLH protocol, whose standard deviations are usually higher than other protocols especially when $\epsilon = 3$ or 4. This is because of the relatively higher utility loss of BLH compared to other protocols especially under high ϵ . Altogether, we arrive at the following take-away message.

Post-processing can improve protection against BIAs by increasing the attacker's inference error, though this benefit is not universal across all PP methods. While no single PP method is universally optimal, Norm-Mul and Norm-Cut are the most consistent in exceeding the No-PP baseline.

5.8. Impacts of PP methods on longitudinal attacks

Next, we evaluate the impacts of PP methods on longitudinal attacks using the LASR metric. Recall that in a longitudinal attack, the adversary observes n perturbed reports $\mathcal{O}_u$ of the same user across

Table 8

BIE results on the Kosarak dataset for different LDP protocols and PP methods. For each protocol and ϵ , the highest BIE (most robust configuration) is highlighted. The PP method with the highest average BIE across all ϵ is in bold and underlined.

GRR							BLH							OLH						
ϵ							ϵ							ϵ						
Method	0.5	1.0	2.0	3.0	4.0	Avg	Method	0.5	1.0	2.0	3.0	4.0	Avg	Method	0.5	1.0	2.0	3.0	4.0	Avg
Base-Pos	0.015	0.009	0.010	0.056	0.053	0.029	Base-Pos	0.010	0.004	0.074	0.182	0.234	0.101	Base-Pos	0.009	0.028	0.030	0.001	0.041	0.022
Norm	0.001	0.005	0.007	0.021	0.021	0.011	Norm	0.013	0.005	0.085	0.054	0.131	0.058	Norm	0.003	0.021	0.025	0.054	0.060	0.033
Norm-Cut	0.018	0.016	0.013	0.012	0.008	0.013	Norm-Cut	0.013	0.045	0.161	0.426	0.649	0.259	Norm-Cut	0.004	0.012	0.111	0.056	0.038	0.044
Norm-Mul	0.016	0.025	0.001	0.017	0.002	0.012	Norm-Mul	0.035	0.010	0.073	0.046	0.637	0.160	Norm-Mul	0.013	0.029	0.086	0.039	0.020	0.037
Norm-Sub	0.001	0.014	0.003	0.036	0.008	0.012	Norm-Sub	0.031	0.001	0.130	0.065	0.524	0.150	Norm-Sub	0.006	0.017	0.083	0.010	0.072	0.038
Power	0.006	0.025	0.008	0.009	0.001	0.010	Power	0.007	0.049	0.122	0.061	0.319	0.112	Power	0.008	0.023	0.094	0.002	0.071	0.040
Power-NS	0.036	0.003	0.005	0.053	0.008	0.021	Power-NS	0.005	0.007	0.114	0.041	0.018	0.037	Power-NS	0.006	0.019	0.056	0.074	0.052	0.041
Avg-PP	0.013	0.014	0.007	0.029	0.014		Avg-PP	0.016	0.017	0.108	0.125	0.359		Avg-PP	0.007	0.021	0.069	0.034	0.051	
No-PP	0.002	0.004	0.022	0.001	0.003	0.006	No-PP	0.005	0.031	0.087	0.123	0.059	0.061	No-PP	0.014	0.043	0.068	0.038	0.017	0.036
RAPPOR							OUE							SS						
ϵ							ϵ							ϵ						
Method	0.5	1.0	2.0	3.0	4.0	Avg	Method	0.5	1.0	2.0	3.0	4.0	Avg	Method	0.5	1.0	2.0	3.0	4.0	Avg
Base-Pos	0.003	0.033	0.065	0.007	0.033	0.028	Base-Pos	0.060	0.005	0.014	0.006	0.047	0.026	Base-Pos	0.019	0.027	0.078	0.045	0.023	0.038
Norm	0.003	0.001	0.037	0.048	0.080	0.034	Norm	0.006	0.033	0.049	0.013	0.007	0.022	Norm	0.001	0.010	0.082	0.044	0.020	0.031
Norm-Cut	0.011	0.058	0.103	0.059	0.205	0.087	Norm-Cut	0.041	0.066	0.065	0.063	0.047	0.056	Norm-Cut	0.039	0.043	0.071	0.042	0.083	0.056
Norm-Mul	0.018	0.001	0.097	0.033	0.034	0.037	Norm-Mul	0.048	0.038	0.006	0.019	0.009	0.024	Norm-Mul	0.019	0.033	0.041	0.070	0.006	0.034
Norm-Sub	0.025	0.011	0.040	0.030	0.136	0.048	Norm-Sub	0.059	0.032	0.030	0.030	0.030	0.036	Norm-Sub	0.028	0.034	0.060	0.020	0.003	0.029
Power	0.002	0.045	0.104	0.041	0.064	0.051	Power	0.049	0.006	0.075	0.031	0.069	0.046	Power	0.012	0.001	0.064	0.090	0.001	0.034
Power-NS	0.011	0.005	0.071	0.006	0.121	0.043	Power-NS	0.037	0.054	0.015	0.001	0.033	0.028	Power-NS	0.026	0.047	0.049	0.065	0.001	0.038
Avg-PP	0.010	0.022	0.074	0.032	0.096		Avg-PP	0.043	0.033	0.036	0.023	0.035		Avg-PP	0.021	0.028	0.064	0.054	0.020	
No-PP	0.004	0.034	0.014	0.014	0.044	0.022	No-PP	0.025	0.003	0.035	0.115	0.042	0.044	No-PP	0.002	0.005	0.056	0.063	0.007	0.026

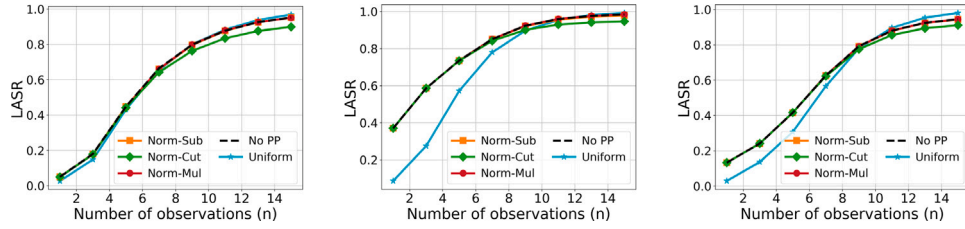


Fig. 7. Longitudinal Attack Success Rate (LASR) of different PP methods under varying numbers of observations n , using BLH and $\epsilon = 4$. From left to right: Adult, Kosarak, BMS-POS.

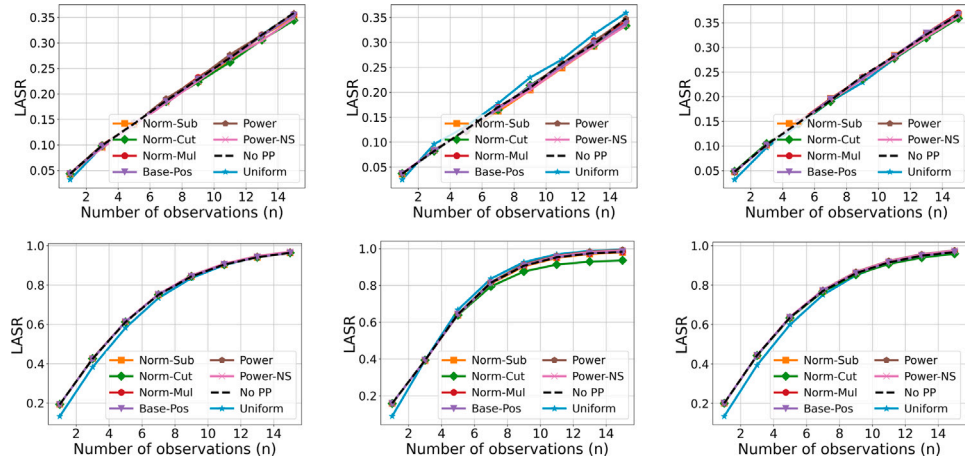


Fig. 8. Longitudinal Attack Success Rate (LASR) of different PP methods under varying numbers of observations n . Top row: Adult dataset with $\epsilon = 1$; bottom row: BMS-POS dataset with $\epsilon = 3$. From left to right: OUE, RAPPOR, SS.

multiple collection rounds and predicts the user's true value via the Bayes rule (Eq. (26)), where the prior $\Pr[v]$ is derived from the post-processed estimate $\tilde{f}(v)$ when PP is used. A higher LASR indicates a more successful attack, i.e., weaker privacy. In Fig. 7, we report results on the three real-world datasets (Adult, Kosarak, BMS-POS) using BLH at $\epsilon = 4$, while varying the number of observations n . For comparison, we also include the No-PP baseline and a Uniform baseline in which the prior $\Pr[v]$ is set to a uniform distribution (i.e., it carries no frequency information).

We observe that, across all three datasets, LASR increases as the number of observations n grows, since repeated observations of the

same user provide the attacker with more evidence to infer the true value. More notably, the PP methods (Norm-Sub, Norm-Cut, Norm-Mul) and the No-PP baseline are tightly clustered throughout, indicating that PP provides little to no protection against longitudinal attacks. This stands in contrast to MGA and BIA, where PP methods caused substantial robustness benefits. The reason is that the longitudinal prediction in Eq. (26) combines a per-user likelihood term $\Pr[\mathcal{O}_u | v]$, which is driven by the user's repeated reports, with a prior $\Pr[v]$ derived from the estimation result. PP methods reshape only this $\Pr[v]$; they do not alter the per-user perturbed reports that determine the likelihood. As n increases, the likelihood term increasingly dominates

the prediction, so the influence of PP becomes marginal, and the curves converge. This also explains the behavior of the Uniform baseline, which removes all frequency information from $\Pr[v]$. It yields the lowest LASR at small n (where the prior still matters), but converges towards the other curves as n grows and the likelihood dominates. Among the PP methods, Norm-Cut is marginally the most robust against longitudinal attacks, but the gaps between all PP methods and the No-PP baseline are small, meaning that the usage of PP provides limited benefit against longitudinal attacks.

To understand whether these results are specific to the BLH $\epsilon = 4$ setting or whether they can be generalized to different protocols and ϵ values, we repeated the same experiment on OUE, RAPPOR, and SS protocols under different ϵ budgets. The results are reported in Fig. 8. The first row corresponds to the Adult dataset with $\epsilon = 1$, whereas the second row corresponds to BMS-POS dataset with $\epsilon = 3$. Each column corresponds to a different protocol (OUE, RAPPOR, SS). The results confirm that the behavior observed in the previous experiment persists across the additional protocols and the lower ϵ budgets: in every case, the PP methods and the No-PP baseline remain tightly clustered, and the same convergence towards the likelihood-dominated regime is observed as n grows. In our experiments, we also observed that the standard deviations of these LASR values are quite low, e.g., on the Adult dataset and $n = 9$, the mean LASR values across 20 experiment repetitions are within the range 0.20–0.24 for all protocols (consistent with Fig. 8), but the standard deviations are all below 0.008. In other words, the standard deviations of the LASR readings are more than an order of magnitude lower than the readings themselves. Considering this low variability, we arrive at the following take-away message.

Unlike MGA and BIA, PP methods provide little to no protection against longitudinal attacks. Since these attacks are driven by repeated per-user observations that PP does not alter, the choice of PP method has limited impact on attack success, and this impact further diminishes as the number of observations grows.

5.9. Efficiency and correctness of the multi-threaded design

In this section, we evaluate the multi-threaded design of LDP³⁺ along two axes: (i) the execution time speedup it delivers, and (ii) validation that this speedup does not hurt the correctness of estimation results. First, we report the execution time experiments to demonstrate the practical benefits of the Multi-Threading Module described in Section 4.5. We measured the end-to-end execution time of the frequency estimation task while varying the number of threads $\eta \in \{1, 4, 8, 16, 24\}$ using an Intel Core i9-14900 K CPU. We performed this experiment on all three real-world datasets (Adult, Kosarak, and BMS-POS). For each dataset, we report results for five LDP protocols (OLH, BLH, SS, OUE, RAPPOR). To isolate the effect of multi-threading, all other parameters were kept fixed across runs, and each configuration was repeated for 20 repetitions, with the reported execution times averaged across repetitions.

The results shown in Fig. 9 clearly demonstrate the advantage of the multi-threaded design. Across all protocols and on all three datasets, execution times decrease substantially as the number of threads increases. The improvement is most pronounced for the computationally heavier protocols: on Kosarak, the execution time of OLH drops from roughly 275 s with a single thread to about 26 s with 24 threads (a speedup of more than 10 $\times$), and BLH exhibits a similar trend. A comparable pattern is observed on BMS-POS. The lighter protocols (SS, OUE, RAPPOR) also benefit considerably, with multi-threading reducing their execution times to under 12 s on these two datasets. The Adult dataset exhibits the same qualitative trend but at a much smaller absolute scale, owing to its smaller size: OLH and BLH drop from roughly 8.5 s with a single thread to about 1 s with 24 threads, while the lighter protocols (SS, OUE, RAPPOR) decrease from around 3 s to under 1 s.

In all cases, the largest relative gains are obtained early (moving from 1 to 4 and then 8 threads), after which the curves flatten as the speedup becomes bounded by per-thread overhead and the sequential portions of the workload (i.e., reading the dataset and parameters, combining the results of the threads at the end). Throughout different η , the relative ordering of the protocols is preserved, indicating that multi-threading reduces absolute execution times without altering the protocols' relative computational costs.

Second, we demonstrate that the speedups gained by multi-threading preserve correctness, by verifying that the multi-threaded design produces the same frequency estimates as single-threaded execution. Intuitively, since averaging equally sized partitions is mathematically equivalent to estimating over the whole population, the result should not depend on the number of threads η . However, the inherent randomness in LDP alone can cause each execution's result to be different from the others. In order to remove the effect of LDP randomness and purely isolate the effect of the number of threads, we fixed the random seeds on a per-user basis (only for the experiments in this section), so that each user in the population will receive the same perturbation regardless of the thread to which they are assigned. This isolates the effect of η : with per-user seeds fixed, any difference in the resulting frequency estimates is caused by η rather than the inherent LDP randomness.

In our first experiment, we measured the ℓ_1 distances of the frequency estimates under varying $\eta \in \{1, 2, 4, 8, 16\}$. We ran this experiment on the same three real-world datasets (Adult, Kosarak, BMS-POS) at $\epsilon = 1$, using GRR, OUE, and OLH, together with two PP methods (Norm-Mul and Norm-Sub). The results are reported in Table 9. As shown in Table 9, the ℓ_1 errors are identical across all thread counts for every dataset, protocol, and PP method. We report results with four decimal places for readability; any residual difference is attributable to floating point arithmetic rather than the number of threads. This confirms that the multi-threaded design alters only execution time and not the estimation result, validating the correctness of our partition-and-combine parallelization.

To complement the numerical results in Table 9 with a visual confirmation, Fig. 10 shows the estimated frequency vectors obtained under $\eta \in \{1, 2, 4, 8, 16\}$ threads on the Adult dataset at $\epsilon = 1$, for all six LDP protocols. It can be observed that for every protocol, the curves corresponding to different thread counts overlap exactly. This overlap is consistent with the findings in Table 9 and shows that the resulting frequency vectors are the same regardless of the number of threads η . Hence, we confirm that the number of threads η affects only execution time, and not the estimation correctness in LDP³⁺.

5.10. Case study: Geospatial density estimation

To broaden the scope of our experiments and incorporate a real-world task using a large real-world dataset, we conduct a case study on geospatial density estimation using the *Porto* dataset. With over 1.6 million records, Porto is roughly three times larger than BMS-POS. It contains trips of 442 taxis driving in the city of Porto, released as part of the Taxi Service Prediction Competition in ECML-PKDD (Moreira-Matias et al., 2013). The original dataset contains full taxi trips, i.e., trajectories with multiple location readings per trip. We pre-processed it by sampling only one location from each trip and treated them as the current locations of the users. Here, the protected unit is an individual location: each sampled location constitutes one user's value v_u , so the ϵ -LDP guarantee applies at the level of a single location rather than at the level of a taxi. This choice is consistent with the geospatial density estimation task and similar pre-processing steps are used in many prior works for converting trajectory data into single locations prior to experimentation. We only used the locations between longitudes -8.691294 and -8.552009 and latitudes 41.138351 and 41.185935 , corresponding to the city of Porto, resulting in 1,620,157 locations. We discretize the geographic area using a 9×9 grid and

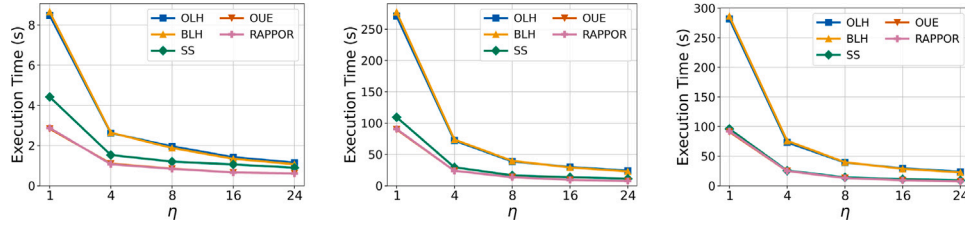


Fig. 9. Execution time (in seconds) of five LDP protocols (OLH, BLH, SS, OUE, RAPPOR) under varying numbers of threads on the Adult, Kosarak, and BMS-POS datasets.

Table 9

Validation of the multi-threaded design of LDP³+ on the Adult, Kosarak, and BMS-POS datasets ($\epsilon = 1$, fixed random seeds are used to reduce the impact of randomness on results). Each cell reports the ℓ_1 distance of the frequency estimates under a given number of threads η .

Dataset	Protocol	PP Method	Number of Threads η				
			$\eta = 1$	$\eta = 2$	$\eta = 4$	$\eta = 8$	$\eta = 16$
Adult	GRR	Norm-Mul	0.0110	0.0110	0.0110	0.0110	0.0110
		Norm-Sub	0.0126	0.0126	0.0126	0.0126	0.0126
	OUE	Norm-Mul	0.0065	0.0065	0.0065	0.0065	0.0065
		Norm-Sub	0.0067	0.0067	0.0067	0.0067	0.0067
	OLH	Norm-Mul	0.0051	0.0051	0.0051	0.0051	0.0051
		Norm-Sub	0.0052	0.0052	0.0052	0.0052	0.0052
Kosarak	GRR	Norm-Mul	0.0040	0.0040	0.0040	0.0040	0.0040
		Norm-Sub	0.0030	0.0030	0.0030	0.0030	0.0030
	OUE	Norm-Mul	0.0013	0.0013	0.0013	0.0013	0.0013
		Norm-Sub	0.0012	0.0012	0.0012	0.0012	0.0012
	OLH	Norm-Mul	0.0014	0.0014	0.0014	0.0014	0.0014
		Norm-Sub	0.0013	0.0013	0.0013	0.0013	0.0013
BMS-POS	GRR	Norm-Mul	0.0038	0.0038	0.0038	0.0038	0.0038
		Norm-Sub	0.0041	0.0041	0.0041	0.0041	0.0041
	OUE	Norm-Mul	0.0017	0.0017	0.0017	0.0017	0.0017
		Norm-Sub	0.0017	0.0017	0.0017	0.0017	0.0017
	OLH	Norm-Mul	0.0016	0.0016	0.0016	0.0016	0.0016
		Norm-Sub	0.0016	0.0016	0.0016	0.0016	0.0016

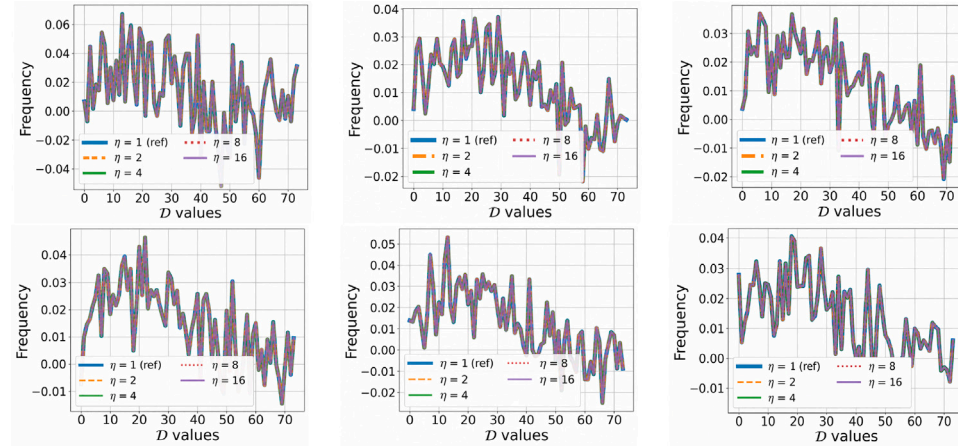


Fig. 10. Estimated frequencies on the Adult dataset ($\epsilon = 1.0$) under $\eta \in \{1, 2, 4, 8, 16\}$ threads, for all six LDP protocols (left to right, top to bottom: GRR, BLH, OLH, RAPPOR, OUE, SS).

assign each user the index of the grid cell in which their location falls. The choice of a 9×9 grid follows prior work [Baloglu et al. \(2024\)](#), which found a uniform grid of this granularity to be near-optimal for the Porto area; this yields a domain of $|D| = 81$ cells over a population of $|P| = 1,620,157$ users.

Utility. We assume a data collector whose task is to perform geospatial density estimation, i.e., find which areas (which grid cells) contain more taxi trips. This can help in multiple ways in the real world, e.g., taxi demand forecasting, urban planning, and profit maximization for taxi drivers. Towards this task, we perform frequency estimation for

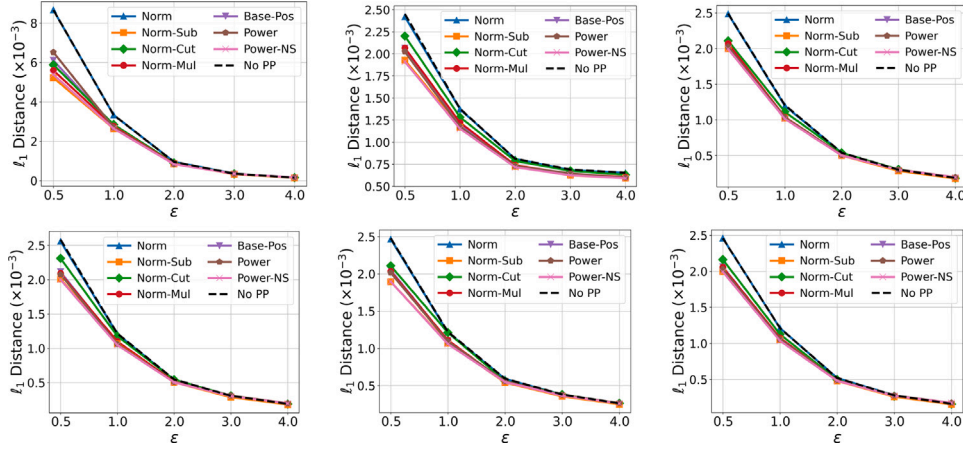


Fig. 11. Average utility loss (measured via ℓ_1 distance) of different PP methods on the Porto dataset under different protocols and ϵ budgets. Top row: GRR, BLH, OLH protocols. Bottom row: OUE, RAPPOR, SS protocols.

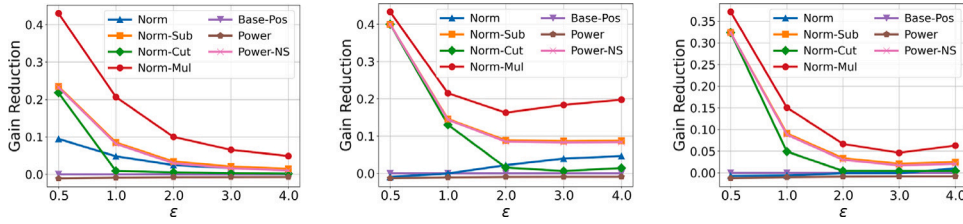


Fig. 12. Gain Reduction (GR) achieved by different PP methods under MGA on the Porto dataset (top- k target set). From left to right: OLH, OUE, RAPPOR protocols.

each cell in D and measure the resulting ℓ_1 errors of LDP estimation + PP. (Note that this setup is very similar to the previous utility experiments we conducted in Section 5.2; the main difference is the new dataset and a different domain.) Fig. 11 contains the results, i.e., ℓ_1 distances of the PP methods across all six protocols and varying ϵ .

The results contain similar trends to those observed on previous datasets. The differences between PP methods are largest when ϵ is small, and the ℓ_1 distances of all methods converge towards the No-PP baseline as ϵ increases, confirming that the benefit of post-processing is more concentrated in the strict privacy regime. In terms of the best method, Norm-Sub, Norm-Mul, and Power-NS attain the lowest ℓ_1 distances at low ϵ , whereas Norm tracks the No-PP baseline closely and offers little improvement. These observations are consistent with our earlier finding that Norm-Sub and Norm-Mul are the most desirable choices for frequency estimation.

Adversarial robustness. Next, we consider an attacker who is aware of the data collector's task and would like to run a poisoning attack using MGA. A successful attack can, for example, lead to erroneous results in taxi demand forecasting or urban planning, causing a waste of allocated resources. We perform a MGA experiment similar to that in Section 5.6 using the most frequent $k = 15$ cells as the attack target. In Fig. 12, we report the Gain Reduction (GR) under each PP method using the OLH, OUE, and RAPPOR protocols. The results again mirror our earlier findings. Norm-Mul is the strongest defender across all three protocols, achieving the highest GR at every ϵ , with Norm-Sub and Power-NS being the next-best defenders. In contrast, Base-Pos and Power provide near-zero GR throughout, consistent with our earlier explanation that per-item methods cannot suppress the inflated mass of a successful poisoning attack. As before, GR is highest at small ϵ and generally decreases as ϵ increases, though OUE again exhibits a less monotonic trend in this regard than OLH and RAPPOR. Overall, Norm-Mul remains the most effective PP method against MGA on this dataset as well.

The case study on the Porto dataset corroborates our earlier findings: Norm-Sub and Norm-Mul remain the most desirable choices for frequency estimation, and Norm-Mul remains the best choice for robustness against MGA. This provides supporting evidence that the main trends continue to hold in the large-scale geospatial setting.

6. Conclusion

In this paper, we designed and developed LDP³⁺, a modular and extensible platform for analyzing the behavior of PP methods in LDP. LDP³⁺ integrates six widely used LDP protocols, seven PP methods, three adversarial attacks, and multiple utility and adversarial evaluation metrics under a multi-threaded execution framework. We conducted extensive experiments using LDP³⁺ and highlighted several take-away messages. Our results show that the effectiveness of PP methods varies depending on the privacy budget ϵ , protocol choice, dataset characteristics, and evaluation metric. Nevertheless, Norm-Sub and Norm-Mul provide strong improvements for frequency estimation tasks, while Norm-Cut performs best for rank preservation tasks. Furthermore, our adversarial analysis reveals that PP methods can indirectly improve the robustness of LDP protocols against attacks such as MGA and BIA. For longitudinal attacks, however, we found that PP methods provide little protection, since these attacks are driven by repeated per-user observations that post-processing does not alter. Overall, our work provides both a practical benchmarking tool (LDP³⁺) and new insights into the role of post-processing in LDP. By enabling systematic evaluation across diverse scenarios, LDP³⁺ can assist future researchers and practitioners in selecting appropriate protocol and PP methods for specific applications.

There are several avenues for future work. First, we plan to maintain and update LDP³⁺ as new LDP protocols and PP methods are proposed in the literature, ensuring our platform remains up to date. Second, we plan to extend the Adversarial Module with a wider range of attack

types and also implement defense mechanisms for further analyses. Finally, we plan to design a graphical user interface (GUI) for LDP³+ and conduct human studies to test how convenient it is for non-experts to use LDP³+ for protocol and PP method selection.

CRedit authorship contribution statement

Alireza Khodaie: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Berkay Kemal Balioglu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M. Emre Gursoy:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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